



Distributed and trustworthy digital twin platform based on blockchain and Web3 technologies

Seyed Mojtaba Hosseini Bamakan^a, Saeed Banaeian Far^{b,*}

^a Department of Management Sciences, Yazd University, Yazd, 89195-741, Iran

^b Blockchain and Metaverse research lab Iran

ARTICLE INFO

Keywords:

DeSci
Fundraising
Intellectual Property
SWOT analysis
Web3 technology

ABSTRACT

The fourth industrial revolution has significantly increased the adoption of Digital Twins (DTs) across various sectors, including intelligent manufacturing, automation, logistics, and medical analysis. Despite substantial progress in deploying DT projects, numerous challenges remain, such as managing distributed data flows, protecting commercial confidentiality, securing intellectual property, and ensuring privacy and security. This research introduces a novel approach to addressing these challenges by leveraging blockchain and Web3 technologies, including non-fungible tokens (NFTs) and Distributed Autonomous Organizations (DAOs). The study aims to develop a distributed, tamper-proof DT cooperation platform that facilitates traceable and trustworthy data sharing while preserving intellectual property rights and enabling decentralized governance. This platform enhances idea and invention ownership, promotes collective decision-making through consensus protocols, and explores innovative solutions like blockchain-based federated learning and efficient DT project fundraising tools. Relying on security models and analysis tools, this study addresses several important security analysis methods/tools/models (e.g., random oracle model, BAN logic, AVISPA tool, and TAMARIN prover) that are known as practical analysis methods. They can strongly prove every claimed security feature of DT projects. The proposed solutions set the stage for future academic and industrial advancements, supported by a comprehensive SWOT (Strongness - Weakness - Opportunity - Threat) analysis, and outline potential future research directions.

1. Introduction

Digital twin (DT), an emerging technology trend, is a digital copy of a physical system, product, and process representing their components and processes. To link the physical and digital world, DTs convert sensor-based collected data into digital information and to better understand who is committing predictions for the future [1]. The concept of DT consists of three parts: the physical product, the digital/virtual product, and the connections between them [2]. DT technology has improved the current theories and knowledge of human beings in building virtual models, exploring and predicting the unknown world, finding better ways to stimulate innovative human ideas, and pursuing optimal progress. DTs are capable of emulating real-world objects and providing natural senses in real-time. To create a high-quality and valuable DT, some steps under the consultation of experts in the related field should be passed carefully [3]. For example, preparing a 2D design, 2D digital design, 3D digital design, high presentation of the digital object, and implementation of all details are some assumed steps related to the DT of an object. DTs enable product managers to use the most

realistic simulations of system details before releasing the real-world version.

The DT technology has moved beyond manufacturing and into merging the Internet of Things (IoT), big data processing techniques, cloud computing, Artificial Intelligence (AI), and other disruptive technologies [4]. According to the Gartner study results, 13% of organizations are already using DTs, while 62% are implementing DTs [5]. It is promoted in numerous applications, including manufacturing, supply chain, smart cities, industrial IoT, services, healthcare, etc. In the last two years, the outbreak of the COVID-19 pandemic has changed production methods and maintenance, which has accelerated the adoption of DTs [6]. Applying DTs to more practical problems requires grabbing newly-presented ideas or technologies such as blockchain and NFT.

NFT is a blockchain-based digital asset representing the control of a unique and irreplaceable asset [7]. Issuers mint NFTs that might be trade-in online marketplaces (e.g., OpenSea, Axie Marketplace, CryptoPunks, Rarible, SuperRare, etc.) as a proof of ownership of physical things or as an investment method for obtaining benefits. Nevertheless, NFTs are not limited to the digital space. They can link each physical

Peer review under responsibility of KeAi Communications Co., Ltd.

* Corresponding author.

E-mail addresses: smhosseini@yazd.ac.ir (S.M. Hosseini Bamakan), saeed.banaeian.far@gmail.com (S. Banaeian Far).

<https://doi.org/10.1016/j.csa.2024.100064>

Received 9 March 2024; Received in revised form 3 June 2024; Accepted 30 June 2024

Available online 5 July 2024

2772-9184/© 2024 The Authors. Publishing Services by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

asset in the real world, as a kind of digital twin, to a digital tamper-proof string allowing users to prove their ownership rights and trade them in digital marketplaces. This connection assured digital twin producers secure control of their underlying assets due to the reliability of blockchain technology. Additionally, NFTs can be accompanied by smart contracts that allow the seller to create conditions on the token-holders' rights, such as royalty payments to the original NFT creator.

NFTs have found their application in a broad spectrum of academic research and industry, such as intellectual property, patents, digital artworks, music, videos, Metaverse-based games, trading cards, tickets, membership proof, and real-world asset ownership. More specifically, scientists from different domains would take advantage of NFTs in academic research such as registering the ownership of the clinical trial results, biotechnology methods and techniques, live-cell images, space and galaxies' images, solo fundraising, workflows, etc. The application of NFTs is not only restricted to the proof of ownership but would also be used to prove the authenticity of intellectual property and patents. NFT properties provide a trusted situation for transfer ownership in the marketplace. Regarding recent investments of well-known companies in Metaverse (e.g., Ripple, eBay Inc, NIKE, Coca-Cola, etc.), it is anticipated that some applications of NFTs, such as using them for loans, intellectual property, personal publication, and regulatory issues for NFTs, will trend in the late 2022 and then 2023.

Despite the various and attractive applications of NFTs, they face some challenges [8], such as mainly being Ethereum-based (ERC721 standard) so which needs the high value of gas price and causes a slow confirmation, the risk of off-chain storage, legal pitfalls and gaps such as money-laundering, tax, etc. that require more attention from the legislative aspects and technical issues, e.g., cross-chain interoperability, and links to real-world oracles.

With the growth of newly-appeared technologies, DTs have attracted a broad spectrum of researchers, including academics, computer scientists, product designers, and industrial managers. Nevertheless, these projects deal with various problems, such as the challenges of big-data-like data flows, the confidentiality of DTs' details of commercial information, massive costs of DT design that require government or academic grants, cooperating DT teams, and multiple security aspects, especially registration and transfer of DT ownership. This study believes that applying newly-proposed technologies and state-of-the-art industrial topics can assist DT teams in solving their challenges [15].

1.1. Motivation and contribution

Digital twins have gained significant momentum in various industries, offering immense potential for enhancing operational efficiency and decision-making processes. However, the growth of digital twins is not without its challenges. One of the primary challenges is the management of Digital Twin big Data Flows [9]. As digital twins generate vast amounts of data from multiple sources, effectively handling, analyzing, and interpreting this data becomes crucial. Additionally, ensuring Commercial Information and IP ownership poses a significant challenge [65]. As digital twins involve collaboration between multiple stakeholders, determining ownership rights and protecting intellectual property become complex tasks. Furthermore, digital twin Fundraising requires innovative approaches to secure funding for the development and implementation of digital twin projects [7]. Establishing distributed collaboration platforms for Twin teams is another challenge, as it necessitates secure and transparent platforms for seamless data sharing and collaboration among multiple entities [16]. Lastly, ensuring Security and Privacy in digital twin projects is paramount, considering the sensitivity and confidentiality of the data involved. Overcoming these challenges will require a multidisciplinary approach, encompassing technological advancements, legal frameworks, and standardized protocols, to unlock the full potential of digital twins and enable their widespread adoption in various domains.

This study relies on blockchain and its deviated technologies, networking techniques, and cryptographic tools to provide solutions for the challenges mentioned earlier. Therefore, the main contributions of this paper are categorized into three items below:

- **Blockchain-based technologies:** Blockchain is selected as the principal technology that provides numerous attractive features, such as immutability, autonomy, and open-source. DTs would be immutable in their design details and programming codes if presented as NFTs. Moreover, they are assumed to be salable blockchain-based assets, and DT project(s) owners can prove project ownership and keep the intellectual property. DAOs and Metaverse, derivative from blockchain technology, provide autonomy for DT projects without a central authority. This enables developers to apply DTs in autonomous computer networks (e.g., DAOs) or virtual AI-based environments (e.g., Metaverse) to provide natural senses for users/consumers. Consequently, DT owners can sell/purchase/share their projects fairly and reliably.
- **Networking techniques:** Several networking techniques can significantly assist in solving DT project challenges. Edge and fog computing help the remotely-working DT teams to process data efficiently with lower latency. Parallel computing enables collaborating DT teams to proceed with their projects individually and merge at the end of project deadlines [70,71]. By relying on the two mentioned examples, sharding techniques provide more velocity and scalability for DT teams to proceed with scattered projects.
- **Cryptographic tools:** Regarding the flexibility of cryptographic schemes and tools, several properties can be provided by using those mentioned tools in such a way that they are also compatible with DT teams' demands and projects. Firstly, adopting Identity-based Encryption (IBE) for twin teams in blockchain-based systems is recommended, providing confidentiality for DT details without needing a central authority like the Public Key Infrastructure (PKI) or a certificate issuer [72,76]. This advances the current state-of-the-art by offering a more secure and efficient encryption method. Furthermore, the project proposes the implementation of self-judgment data authentication protocols to ensure the integrity of DT details. By enabling twin teams to detect any intrusion and changes in DT details individually, even without a central authority or judge, this approach enhances the state-of-the-art by providing a reliable and decentralized method for maintaining data integrity.

In addition, the project suggests the utilization of Zero-Knowledge Proofs (ZKPs) to connect the birth certificate of DT to NFTs [10]. In fact, ZKP provides evidence of ownership for DT project owners with revealing no bit of information about the DT design and code. This enables users to prove their ownership of DT, as linked to NFTs, without revealing sensitive DT details or risking any commercial information leakage. This advancement in technology ensures privacy while establishing a secure and transparent connection between DT and NFT ownership, thus surpassing the current state-of-the-art in terms of privacy and data protection.

Secret sharing protocol (SSP) enables DT teams to work on a private DT project that requires the presence of all groups (e.g., owners or developers can reconstruct a private DT if they all exist) [10]. As another cryptographic scheme, threshold schemes [14] make a democratic environment so that a project is run if a certain percentage of owners exist in project execution.

This study offers solutions for five DT project challenges based on the above descriptions and Fig. 1.

1.2. Outlines

The rest of this study is organized as follows: Section 2 reviews this study's preliminary and background. As the main section, Section 3 discusses five challenges of DT projects addressed in [1] and provides solutions using blockchain-based technologies (e.g., NFTs, DAO, Metaverse,

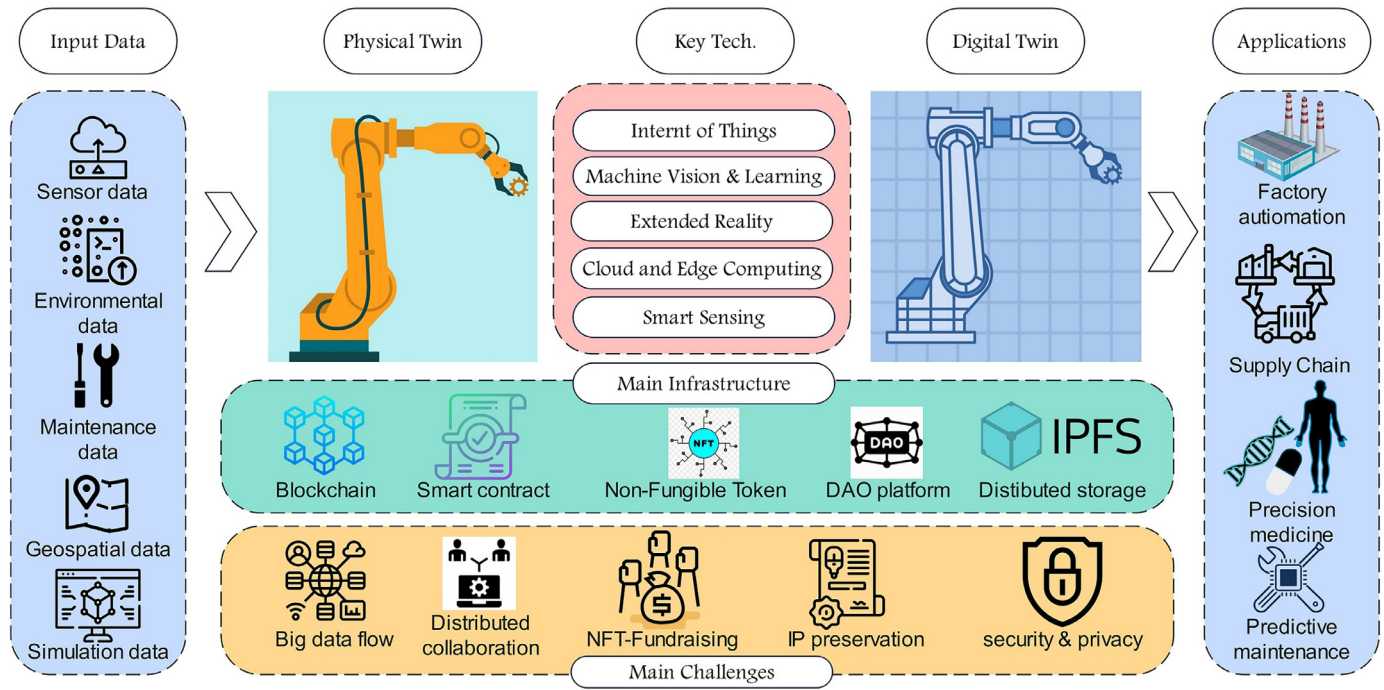


Fig. 1. An overview of Digital Twin platforms and their challenges discussed in this study

etc.) and state-of-the-art techniques and tools. Section 4 presents a brief discussion on offered solutions. Finally, Section 5 concludes this study and identifies several future horizons. Fig. 2 depicts the mentioned outlines in detail.

Table 1 shows the used acronyms and notations in this paper.

2. Background technologies

This section identifies the preliminaries and background of this study; Section 1.1 presents the motivation of this study and briefly reviews the paper contribution.

2.1. Digital twins

As a highly accurate simulation that considers aspects of physical behaves, the concept of DT was first proposed by NASA in the 70s to enhance product maintenance [17,18]. Several technologies have merged with DTs and improved this valuable part of the industry in such a way that DTs now play one of the main roles in the industry. This section first identifies the critical key technologies of DTs and then addresses differences by simulation, their applications, and some general challenges of DT projects. The relations of the mentioned items are summarized and depicted in Fig. 3.

2.1.1. Key Technologies

DTs are not single technologies, and they get much help from other fields of science. In fact, they are designed based on several technologies and sciences empower them and cause them to present more accuracy [21]. In the following, some important of the mentioned technologies are addressed briefly.

1. **IoT:** As one of the main parts of each IoT network, connected wireless sensors are excellent selections for collecting environmental data. They can be considered data acquirer for processing by DTs.
2. **Smart sensing and intelligent perception:** IoT sensors are essential components to support sensing, data gathering, and measurement; and AI technology brings intelligent perception and argument. DTs rely on the combination of these two to achieve highly accurate presentation and output(s).

3. **Machine vision:** It is an engineering tool for digital devices, computer networks, and industrial control. Machine vision processes received data (e.g., images, videos, and even environmental information) to provide AI-assisted visual senses.
4. **Extended reality:** A combination of the above items provides an attractive, user-friendly achievement. Relying on intelligent sensing and AI-based machine vision, extended reality brings a visual sense for human-atop DT projects.
5. **Cloud computing:** Cloud computing helps low-power and/or low-memory devices assign their calculation and/or outsource their data. As components that process data and generate outputs, DTs, executed on weak computers, should be supported by robust technology, such as cloud computing.
6. **Edge computing:** To reduce latency and global network traffic, edges process data in closer places to users and return results to them. DTs that need fast responses benefit from edge computing.

2.1.2. Digital twin and simulation

Although DTs have some similarities with computer-based simulations, they are not the same as simulations. As the first aspect, DT is a highly accurate version or the twin of a physical thing appearing on a computer [17]. In fact, the DT is simulated in the twin of the physical and active environment with all properties [19,20], so that DT involves several simulations and two-sides data flows.

Therefore, it can be said that simulation supports narrow scopes and is suitable for static and primary designs; DT provides comprehensive aspects and is generally used for the business version of products. It is resulted that the main differences between DT with simulation are (i) real-time simulation and relation with the physical environments, (ii) enhancing product designs for business and market, and (iii) optimization in the real world that helps with diagnosis.

2.1.3. General and state-of-the-art application

Although DTs are well-known concepts, they are improved day-by-days and applied in new fields of science and technology [21–23]. Some state-of-the-art DTs' applications are listed in the following:

1. Introduction	1.1. Motivation and contribution
	1.2. Outlines
2. Background technologies	2.1. Digital twins
	2.2. Blockchain and related technologies
	2.3. Relations of technologies and their interoperability
3. Reviewing the main DTs challenges and their possible solutions	3.1. Data Flows
	3.2. Commercial information
	3.3. Fundraising
	3.4. Distributed collaboration platforms
	3.5. Security and privacy
4. Discussion	
5. Conclusion	

Fig. 2. Paper organization

- *Building information modeling (BIM)*: Although DTs and BIMs are traditional concepts, they are still known as state-of-the-art technologies. Embedding DTs in BID designs is an smart idea that makes BIMs more applicable and realistic [24]. It is noted that DTs are mainly different from BIMs, but they also have some similarities. DTs model people's/neighbors' interactions/behaviors with the building's devices, and BIMs support the building's design and construction.
- *Automatic data detection*: This application of DTs mainly uses AI to detect critical data or anomalies in data flows and processes [25]. Automatic data detection assists product owners to find anomalies in trial versions of sensitive products before marketing.
- *Medicine*: Passing all experiments successfully is strongly required for making artificial hearts and people's synthetic body parts, and highly accurate DTs are the best choice for prototype tests [26–30]. This application can be considered the most sensitive application of DTs since it relates to people's lives.
- *Factory automation*: DT is one of the main parts of the revolution of Industry 4.0 and is used in factories to make process monitoring easier [31]. DTs provide real-time monitoring and data tracking for operators, and they alert managers before the devices fail.
- *Planning and control of manufacturing systems*: Using DTs is a way to predict manufacturing systems crashing and devices failing [32].

Based on AI, they make manufacturing control systems smarter and logistic management more efficient.

- *Big data*: DTs are considered both receivers and creators of big data [33]. They enhance/process received big data using AI and generate big data flows similar to their physical versions in the real world.
- *Cloud-based cyber-physical systems*: As DTs represent physical and virtual worlds, they are able to simulate the behavior of real adversaries around clouds [34]. It is suitable for cloud-based cyber-physical systems and cyber officers to apply DTs in finding and preventing cyber threats.
- *Space traveling*: Traveling among planets is an exciting dream; however, in real/physical environments, walking on other planets, operations outside the Earth's atmosphere, space missions, etc., are high-risk projects [35]. DTs significantly assist shuttle designers and astronauts in sensing natural feel from beyond the Earth. Moreover, they will be very cost-efficient and easy-accessible for whom want to experience space travel.
- *Performance monitoring*: High-efficient manufacturing is part of the revolution of Industry 4.0, and performance monitoring should always be available [36]. DTs bring this property for operators that monitor industrial production processes (based on human, digital, or machine vision) and enable them to help in increasing performance.

Table 1
The list of acronyms and notations

Acronym/Notation	Description
3D	Three-Dimensional
$\mathcal{A}$	A PPT adversary
AVISPA	Automated Validation of Internet Security Protocols and Applications
AI	Artificial Intelligence
$ADV_{X,\mathcal{A}}$	The advantage of $\mathcal{A}$ in X
BAN	Burrows, Abadi, and Needham
BIM	Building Information Modeling
$\mathcal{C}$	Cipher space
DAO	Distributed Autonomous Organizations
DT	Digital Twin
DeFi	Decentralized Finance
DeSci	Decentralized Science
ERC-***	Ethereum Request for Comments- <i>No.</i>
$EX_{X,\mathcal{A}}^{real}$	Experiment causes the real output of X
$EX_{S,\mathcal{A}}^{ideal}$	Experiment causes the ideal output of X
GPU	Graphics Processing Unit
HLSL	High-Level Protocol Specification Language
IBE	Identity-based Encryption
ICO	Initial Coin Offering
IF	Intermediate Format
IIoT	Industrial Internet of Things
IoT	Internet of Things
ML	Machine Learning
MoT	Metaverse of Everything
MPC	Multi-Party Computation
n	The number of all users
NFT	Non-Fungible Tokens
PE	Private Equity
PKI	Public Key Infrastructure
PoE	Proof-of-Existence
PPT	Probabilistic Polynomial-Time
RBAC	Role-Based Access Control
RO	Random Oracle
S	Random oracle simulator modeled by $h(\cdot)$
SSP	Secret-Sharing Protocol
SWOT	Strongness - Weakness - Opportunity - Threat
STO	Security Token Offerings
t	The threshold number of users should be cooperate
TAR	Tactical Augmented Reality
UI	User Interface
UX	User Experience
VC	Venture Capital
VPN	Virtual Private Network
Web3	The third generation of Internet (blockchain-based)
WSL	Windows Subsystem for Linux
zk-SNARK	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge
ZKP	Zero-Knowledge Protocol

- *Cyber-physical threat detection*: In addition to purely-digital threats to systems, security officers should also prevent physical threats [37,38]; detecting them is a primary issue before preventing them. Therefore, it seems highly-accurate DTs are necessary for cyber-physical threat detection.
- *Data-driven modeling*: Relying on machine learning (ML) and deep learning, DTs are applied to empower data-driven modeling, and AI enables DTs to detect models to increase performance [21]. Therefore, DTs are used in the industry as robust data-driven interfaces for manufacturing.

2.1.4. General challenges

Although this study reviews the details of the five main challenges of DT projects described in [1], this section lists some general challenges of DTs briefly [39] in the following:

- *Data handling*: As virtual versions of physical tools, DTs are able to send/receive data and process them. However, the handling of big data is known as a challenge. Although this challenge is not limited to big data, DTs should be able to handle data in different environments.
- *Fidelity*: Providing a fidelity guarantee between input(s) and output(s) in DTs to support maximum similarity should be considered in DT designing.
- *Simulation and simulation-based optimization*: Computer-based simulation is a good-but-challenging way for optimizing systems and algorithms [19]. Similarly, DT-based optimization is not easy since it is required to consider DT issues in addition to optimization aspects.
- *Global Implementations*: Calling worldwide experts to join and improve the scientific level of DT teams is essential for project owners; in other words, scattered DT teams collaborating is nowadays considered a challenge since global project Implementations and team synchronization is a complex problem.
- *Deep learning with DT*: Deep learning is a well-known and state-of-the-art field of data science. To increase DT project efficiency, deep learning can be very applicable. However, combining deep learning challenges with the hardnesses of DT designing puts more pressure on DT developers.
- *Interoperability with existing softwares*: Each newly-designed DT is programmed via specific codes (programming language) and is specialized for a determined company. Therefore, compatibility and interoperability with existing software are challenges for DT designers.

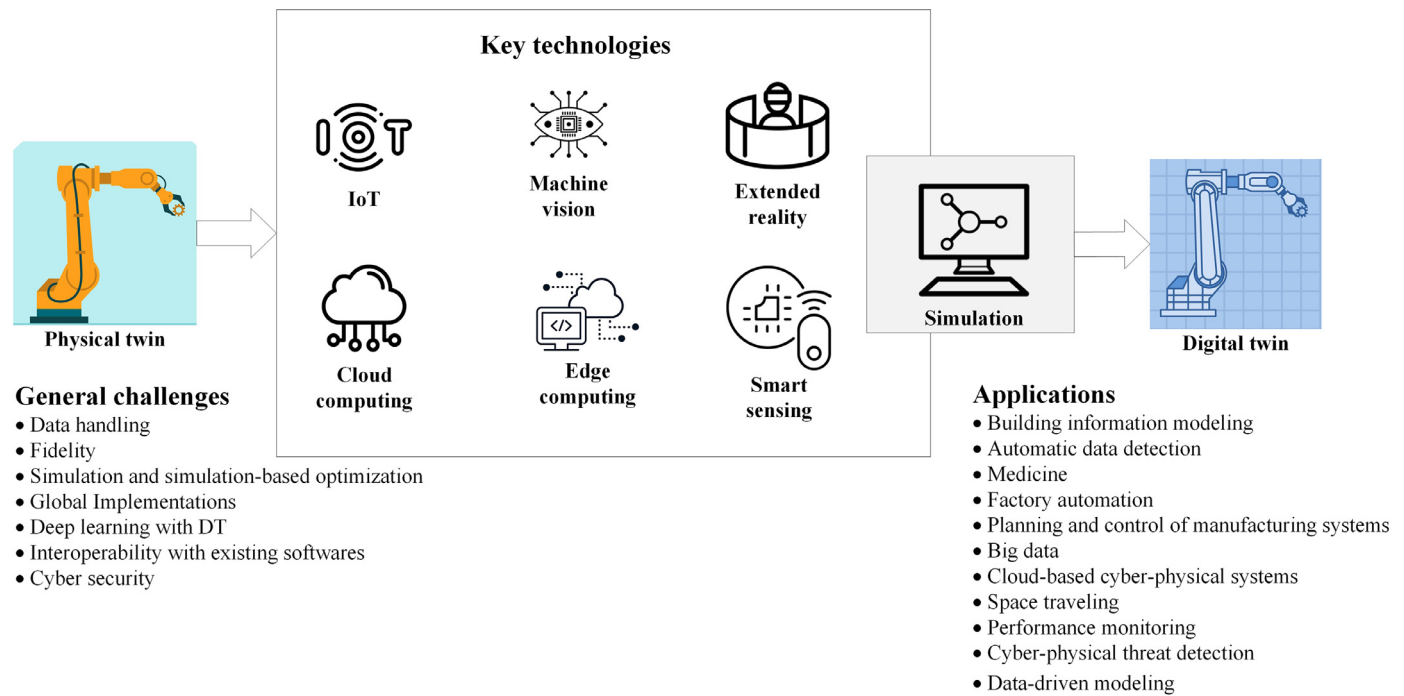


Fig. 3. The summary of Section 2.1

and developers [20]. In fact, the absence of a unified environment, software, and programming language is sensed; and launching DTs with different languages in such a way that are compatible with all projects and softwares is hard.

- **Cyber security:** This item is not only for DT projects and is all times considered for all systems and projects; DT projects are not far from this fact [37,38], and most security and privacy aspects should be assumed in them (for details, refer to [40,41] and Section 3.5).

2.2. Blockchain and related technologies

This study believes that the blockchain technology and its derivative technologies could provide much assistance to solve DT challenges. Therefore, this section first presents the definition of blockchain and then addresses blockchain-based technologies as the background of proposed solutions. The relations of blockchain and its derivative technologies are summarized and depicted in Fig. 4.

2.2.1. Blockchain

Blockchain, one of the most transformative and disruptive technologies of the last decade, was practically introduced in 2008 [42]. This innovation is characterized as a public-access and distributed ledger system, which functions as an intrinsic state machine capable of processing and responding to complex queries. Blockchain technology facilitates the execution of open-source code in a decentralized manner, known as smart contracts [43]. This capability fundamentally alters the landscape of software deployment, enabling greater transparency, efficiency, and automation across various sectors. Furthermore, the blockchain paradigm ensures immutability and heightened security for recorded data and transactions, making it a robust solution for safeguarding digital information.

Following the establishment of a distributed consensus, data is compiled into a block, which is then poised for recording on the blockchain [44]. The integrity of the blockchain is maintained through the use of one-way hash functions, which cryptographically link each block to its predecessor, forming an unbroken chain that traces back to the genesis block. The cryptographic security provided by these hash functions is paramount, as it underpins the entire blockchain's resilience against

tampering and fraud. This ensures that once data is recorded within a block, it remains immutable and verifiable, thereby enhancing the trust and reliability of the blockchain system.

2.2.2. Tokens and non-fungible tokens

For the first few years, blockchain technology was used as the distributed public ledger of cryptocurrencies (Bitcoin is the first one [42]); developers launched their cryptocurrencies on their blockchains, but blockchain owners welcome other developers to release other cryptocurrencies on existing blockchain as tokens. In the following, three types of blockchain-based tokens are listed.

- **Token:** Tokens are a subset of altcoins that do not possess their own independent blockchains; instead, they are created and operate on pre-existing blockchain infrastructures [45]. Their functionality and reliability are intrinsically tied to the performance and stability of their underlying blockchain. Among the various blockchains, Ethereum stands out as the preeminent and most renowned platform for token development, offering robust and versatile frameworks for decentralized applications (dApps) and token issuance.
- **NFT:** In 2017, the concept of NFTs was introduced, grounded in the ERC-721 standard [46]. NFTs are distinguished by their key attributes: immutability, uniqueness, and non-interchangeability. These characteristics ensure that each NFT is a singular, irreplaceable entity, thus enabling the representation of ownership and authenticity for digital and physical assets in a manner that is both secure and verifiable. The advent of NFTs has significantly impacted various domains, including art, gaming, and real estate, by providing a new paradigm for digital ownership.
- **Dynamic NFT (NFT2.0):** Dynamic NFTs represent an evolution of traditional NFTs, featuring metadata that can be altered based on specific conditions or events [47,48]. This dynamic nature allows for on-demand modifications or updates linked to the underlying NFT, such as progressing through levels in a game and acquiring new skills, enhancing the ranking in a blockchain-based market [49], or customizing a virtual office in the Metaverse. Consequently, dynamic NFTs eliminate the necessity to create or purchase a new NFT for

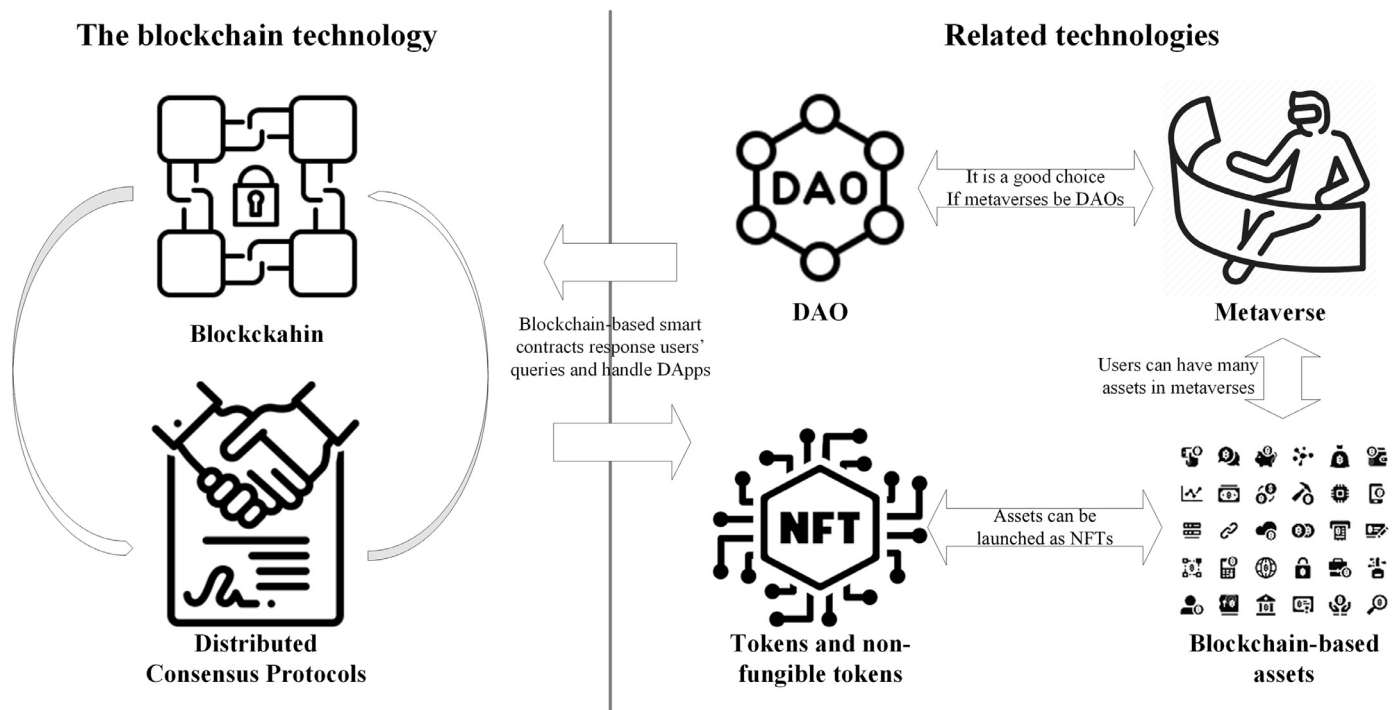


Fig. 4. The summary of Section 2.2

each update, providing a more efficient and flexible mechanism for reflecting changes and developments in the associated digital assets.

2.2.3. Blockchain-based assets

There are several types of blockchain-based assets. Some of them related to this study will now be listed below:

- **Cryptocurrencies, tokens, and NFTs:** These assets represent some of the most widely recognized and utilized values recorded on public blockchains. They serve multiple purposes, including being mediums of exchange, commodities for trading and investment, and methods of payment for a variety of services. Their decentralized nature and cryptographic security make them fundamental components of the burgeoning digital economy.
- **Stakes:** Blockchain-based staking constitutes an investment strategy where users lock their assets into a blockchain protocol to support network operations, such as transaction validation, in return for rewards. These staked assets, or stakes, not only contribute to the network's security and efficiency but also serve as a form of passive income for the asset holders.
- **Securities:** The use of blockchain technology to record securities, such as documents or shares of companies, provides a robust mechanism to ensure their immutability and protection against unauthorized alterations or destruction. This application enhances the transparency, traceability, and overall trust in the management and trading of securities.
- **Data sets:** Researchers and data scientists leverage public blockchains to share statistical data sets openly, thereby promoting transparency, reproducibility, and collaboration in scientific research. The decentralized and immutable nature of blockchain ensures that these data sets remain accessible, tamper-proof, and verifiable by any interested parties.
- **DTs:** DTs, as a central focus of this study, are represented as NFTs and are considered valuable blockchain-based assets [50,51]. DTs encapsulate the digital representation of physical or logical entities, enabling real-time data synchronization and analysis. The subsequent sections will provide a more detailed exploration of the significance and applications of DTs within the context of blockchain technology.

2.2.4. Distributed autonomous organization

As a definition, DAO is an interactive, computer-based community that operates autonomously, devoid of centralized control by any individual or group(s) of authorities/privileged entities [52]. Within a few years, the concept of DAO has gained significant traction and has captivated a broad and diverse audience worldwide. Numerous DAOs, such as UniSwap, MakerDAO, LidoDAO, AAVE, Curve, Dash, KyberDAO, and The LAO, predominantly on the Ethereum and Tron blockchains, have been established. These organizations exemplify the potential of DAOs to revolutionize traditional governance models by promoting transparency, inclusivity, and decentralized decision-making.

Democracy-based communities and governments are increasingly recognizing the potential of DAOs to complement and enhance their policies and practices [53]. This acceptance has led to the formulation of several DAO-based concepts aimed at improving various aspects of societal governance [12,54]. One of the most prominent applications of DAOs is in the realm of decentralized exchanges (DEX), which facilitate peer-to-peer trading of digital assets without the need for intermediaries. The applicability of DAOs extends far beyond this, encompassing numerous sectors and functions.

The reliance on blockchain technology is anticipated to further expand the applicability and influence of DAOs, fostering greater collaboration among dispersed researchers and scientists across the globe. By leveraging the decentralized and secure nature of blockchain, DAOs have the potential to create cohesive, collaborative environments that transcend geographical and institutional boundaries, thereby driving innovation and progress in various fields.

2.2.5. Metaverse

Relying on blockchain technology [55], a Web3-based digital world empowered by AI technology, named the Metaverse, was defined [56]. Within the Metaverse, every virtual entity, whether a person or an object, can possess unique identities and be distinguishable from others. Some studies advocate for the creation and assignment of NFTs for each entity or component within the Metaverse [46,57]. In these studies, each newly-entered user or component of the Metaverse is recognized by a

unique ERC – 721-based identity [65], thereby ensuring distinct and immutable digital identities.

The Metaverse, often described as an environment for people's "second lives," mirrors real-world applications and facilitates the fulfilment of various daily needs on a global scale. This includes sectors such as healthcare services, military training using tactical augmented reality (TAR), real estate, education, fashion, and gaming [58,61]. The immersive and interactive nature of the Metaverse allows for innovative approaches to these fields, enhancing user experience and operational efficiency. Moreover, several governments have shown a keen interest in adopting Metaverse technology [59,60]. Notably, the government of North Korea is among the early adopters of Metaverse technology, exploring its potential for various administrative and societal applications.

2.3. Relations of technologies and their interoperability

Although the above technologies seem exciting and applicable, there are several ambiguities to apply them to improve DT projects. In this regard, we now provide a brief description to justify merging the aforementioned technologies (detailed reasons are discussed in Section 3).

2.3.1. How can blockchain technology empower DT projects?

Providing exciting features by blockchain technology can improve many DT projects in several aspects if applied [50,51]. Some important are listed in the following.

1. **Data integrity:** Ensuring the integrity of collected data from sensors and equipment since it is a tamper-proof ledger, and providing a secure database for storing and sharing DT data, reducing the risk of data breaches.
2. **Real-time data synchronization:** The blockchain's distributed nature enables real-time data sharing among multiple DT project managers. Moreover, it allows DT project developers to access the latest information and make informed decisions promptly.
3. **Transparency and traceability:** All recorded DT projects' data on a public or consortium blockchain are visible to all participants. Therefore, transparency and traceability enable DT project developers, scattered around the world, to track data sources and activities related to the digital twin.
4. **Reduced costs and improved efficiency:** Blockchain technology eliminates the need for intermediaries and reduces DT projects' administrative overhead associated with traditional data management systems and their autonomy. This feature streamlines data exchange and collaboration, leading to cost savings and increased efficiency.
5. **Data decentralization and redundancy:** As an Internet-based distributed ledger, blockchain distributes DT data across a network of nodes or DT teams to provide a database with less susceptible to single points of failure. The data decentralization also ensures data redundancy and availability, even if certain nodes experience outages.
6. **Predictive Analysis:** Recorded DT data on blockchain can be used to train ML models for predictive analysis. In this regard, ML and AI enable DT project managers to identify patterns and anomalies in the data for excellent proactive maintenance and optimization.
7. **Data sharing and worldwide collaboration:** A common database provides a secure and controlled platform for DT managers and developers to share data and reliable collaboration. Therefore, this property facilitates knowledge transfer and innovation across different disciplines and organizations.

2.3.2. How can DTs improve Metaverse applications?

Metaverse-based DT teams (common workspace on Metaverse) provide excellent and cost-efficient collaboration for worldwide DT teams. We categorize them into two aspects, virtual representations of real-world objects [57,61] and user experience (UX) [77–79], listed in the following:

1. **Perfect virtual representations:** DTs play a crucial role in metaverses by creating virtual representations of real-world objects, environments, and systems [57,61]. Several main applications are below.
 - **Virtual prototyping and simulation:** DTs allow users to create and test virtual prototypes of products, buildings, and processes before building them in the real world. This can save time, money, and resources, and reduce the risk of costly mistakes.
 - **Remote monitoring and control:** DTs can be connected to sensors and devices in the real world, allowing users to remotely monitor and control physical assets. This can improve efficiency, reduce downtime, and enable predictive maintenance.
 - **Immersive training and education:** DTs can create realistic and immersive training environments, allowing users to practice skills and learn in a safe and controlled setting.
 - **Data analytics and optimization:** DTs collect and analyze data from connected devices and sensors. This data can be used to optimize processes, identify inefficiencies, and make data-driven decisions.
 - **collaboration and communication:** DTs provide a shared virtual space where users can collaborate and communicate with each other. This can facilitate design reviews, project planning, and knowledge sharing.
2. **UX:** In addition to providing a perfect representation of real-world objects and components, Metaverse-based workspaces make an attractive UX [77–79] which has the following properties.
 - **Enhanced immersion and realism:** DTs can create virtual environments that are highly detailed and realistic, enhancing the user's sense of immersion in the metaverse.
 - **Personalized experiences:** DTs can be tailored to individual user preferences and needs, providing personalized experiences that are more engaging and relevant.
 - **Interactive and collaborative Environments:** DTs enable users to interact with virtual objects and environments, and collaborate with others in real-time. Therefore, this feature creates a more interactive and social experience.
 - **Data-driven optimization:** Data collected from DTs can be used to analyze user behaviour and optimize the metaverse experience. This can lead to improved navigation, content discovery, and overall user satisfaction.
 - **Virtual try-ons and simulations:** DTs can be used to create virtual try-ons and simulations, allowing users to experience products and services before making a purchase or commitment to improve decision-making and reduce the risk of dissatisfaction.

2.3.3. How can NFTs and DAOs increase DT project reliability?

This section first describes, in separate parts, how NFTs and DAO communities increase DT projects' reliability. It then discusses the benefits of the common usage of NFTs and DAOs in DT projects.

1. **NFTs:** The NFTs' uniqueness provides many excellent features [7,65] that can empower DT projects. Three important are listed in the following.
 - **Proof of ownership and authenticity:** NFTs can be used to represent ownership of DTs, ensuring that they are genuine and not counterfeited.
 - **Data provenance and traceability:** NFTs can store metadata about the DT's creation, ownership history, and data sources, in such a way that provides transparency and traceability for increasing trust in the data.
 - **Secure data sharing:** NFTs can be used to control access to sensitive data associated with DTs. Therefore, this NFT feature ensures that only authorized parties can access and use the data, improving data security and privacy.
2. **DAO:** DAOs involve several parties, teams, and even governments who can join to use DAOs' applications, monitoring, or participate in

voting [12,66]. DT teams are not far from this fact, and they can collaborate in common international DT projects as DAOs. Three benefits of this issue are addressed in the following.

- *Decentralized governance*: DAOs allow DT project managers and developers to participate in decision-making regarding the development and use of projects. This ensures transparency, accountability, and DT team involvement.
- *Team-driven validation*: DT teams, as DAO communities, can establish standards and best practices for DT creation and use, which assists in ensuring the quality, reliability, and interoperability of digital twins.
- *Incentivized contributions*: DAOs can reward community members for contributing to the development and improvement of digital twins. This incentivizes participation and ensures that the digital twins are constantly evolving and improving.

All in all, combined benefits of NFTs and DAO communities as the background technologies of DT projects provides the following enhancements:

1. *Enhanced data security and privacy*: NFTs and DAOs work together to ensure that projects' data associated with DTs is secure, private, and controlled by the community.
2. *Improved data quality and reliability*: DAO communities (DT teams) can establish standards and incentivize contributions to improve the quality and reliability of DT projects.
3. *Increased trust and transparency*: NFTs and DAOs provide transparency and traceability, allowing DT project managers to trust the authenticity and integrity of digital twins.
4. *Team-driven innovation*: DAOs foster community involvement and incentivize innovation, leading to the development of more advanced and reliable digital twins.

In conclusion, by leveraging NFTs and DAO communities, DT projects can increase their reliability, transparency, and community engagement, ultimately leading to more trustworthy and valuable digital representations of the physical world.

3. Reviewing the main DTs challenges and their possible solutions

This section discusses the aforementioned DTs challenges and presents solutions based on blockchain, its derivative technologies, and advanced cryptographic tools. The summary of this section is illustrated in Fig. 5.

3.1. Data Flows

DTs collect several unstructured data from distributed sensors and other instruments and analyze them [9]. The data flow of DTs that are received from thousands of distributed sensors of temperature, pressure, speed, etc., which are considered a big-data flow with the properties of huge velocity, variety, and volume that are difficult to aggregate and merge.

3.1.1. Data analytic methods

Some DTs collect/generate high velocity, variety, and veracity data. These types of data, generally known as big data [62], are required to be analyzed and taxonomy and considered a challenge for DT projects [39]. Traditional data analytic methods often fail to handle the complexity and scale of DT data flows. Therefore, DT developers should provide a robust compatible-to-DT data analytic method in their projects designed by a data scientist expert.

1. *ML and AI*: Advanced ML and AI techniques are critical for analyzing large datasets in real-time [80–82]. ML algorithms can be trained to identify patterns, predict anomalies, and optimize processes based on historical and real-time data. Deep learning, a subset of ML, is

particularly useful for handling unstructured data like images and videos collected by DTs.

2. *Stream Processing Frameworks*: Frameworks such as Apache Kafka¹, Apache Flink², and Apache Storm³ are essential for real-time data processing. These tools enable the continuous ingestion, processing, and analysis of data streams, providing near-instantaneous insights and facilitating real-time decision-making.
3. *Data Fusion Techniques*: Combining data from multiple sources (sensor fusion) enhances the accuracy and reliability of the information provided by DTs [83,84]. Techniques like Kalman filtering and Bayesian data fusion are employed to merge data from various sensors, accounting for uncertainties and providing a coherent view of the physical system.

3.1.2. Edge and fog computing

The combination of blockchain and edge computing has recently emerged as state-of-the-art technology and has become popular in Industry 4.0 and the Industrial Internet of Things (IIoT). Edge computing ensures low-latency services, and blockchain guarantees decentralized methods for ensuring data integrity, trust, and security [73]. These two technologies assist the performance and efficiency of systems that face a vast number of data flows. DT simulators are not far from IIoT systems, and they have the potential to be implemented on a public or private blockchain.

Edge computing and hierarchical architecture are used as solutions for solving the challenges of data flows. Edge computing, as a solution, addresses the challenges of variety and high volume by [85,86]

1. *Data Filtering and Preprocessing*: Edge devices can filter out irrelevant or redundant data, preprocess raw data by normalizing or aggregating it, and perform initial analytics to reduce the volume of data transmitted to central servers.
2. *Data Correction*: Edge nodes can correct corrupted data using error detection and correction algorithms, ensuring data integrity before forwarding it to the cloud or central system.
3. *Hierarchical Data Aggregation*: A hierarchical architecture allows for multi-level data aggregation. Cluster heads aggregate data from local sensors, perform initial processing, and then transmit summarized data to higher layers, reducing the overall data traffic and latency.

In complementing of the edge computing, **Fog computing** extends cloud capabilities closer to the data source, providing additional computational resources and storage [87,88]. This reduces the latency and bandwidth requirements by processing data locally or within the vicinity of the data source

3.1.3. Parallel computing

Using multi-core processors or connecting several computers are methods to speed up data processing (DTs' data flows). Designing and programming calculations to be compatible with parallel computing is a field of computational sciences and an excellent choice for making DTs fast and efficient. However, it is a challenge for DT teams, distributed all over the world, to enable parallel computing in their projects and provide a secure infrastructure to share their universal projects.

1. *Distributed Computing Frameworks*: Frameworks like Apache Hadoop⁴ and Apache Spark⁵ enable the distribution of data processing tasks across multiple nodes in a cluster. These frameworks facilitate the parallel processing of large datasets, significantly speeding up the analysis and improving the efficiency of DTs.

¹ <https://kafka.apache.org/>.

² <https://flink.apache.org/>.

³ <https://storm.apache.org/>.

⁴ <https://hadoop.apache.org/>.

⁵ <https://spark.apache.org/>.

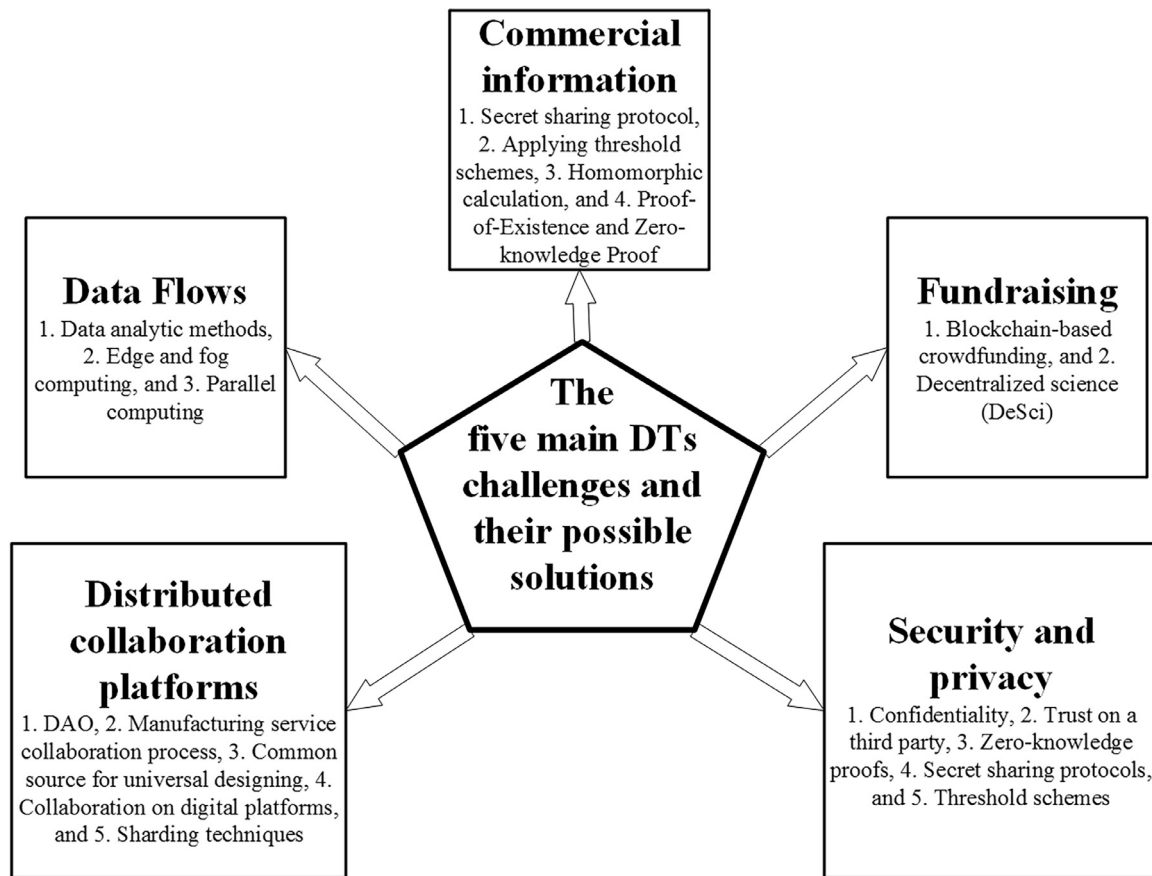


Fig. 5. The summary of Section 3

2. *Graphics Processing Unit (GPU)-Accelerated Computing*: GPUs offer massive parallelism and are highly effective for computationally intensive tasks such as deep learning and image processing [89,90]. Leveraging GPU-accelerated computing can dramatically enhance the performance of DTs, especially for tasks requiring high throughput and real-time processing.
3. *Federated Learning*: This approach allows for training machine learning models across multiple decentralized devices while keeping the data local [91–93]. Federated learning enhances data privacy and security and is particularly useful for collaborative DT projects where data sharing is restricted or impractical.
4. *Secure Multi-Party Computation (MPC)*: Ensuring secure and private computation in distributed environments is crucial. Techniques such as secure MPC allow multiple parties to jointly compute a function over their inputs while keeping those inputs private. This is especially valuable in DT projects involving sensitive data from various stakeholders.

By integrating these advanced techniques and frameworks, DT projects can overcome the challenges posed by high-velocity, high-variety, and high-volume data flows, ensuring efficient, secure, and real-time data processing and analysis.

3.2. Commercial information

Due to the complexity of DT designs, participating teams and companies insist on protecting their products' commercial information from their competitors. For example, various countries consider DT production as a formidable competitive advantage and restrict sharing the DT details with others [1]. To protect commercial information and prevent data leakage, applying data encryption algorithms, SSPs [10], and

threshold validity methods [14] should be used to secure digital interconnections and correspondences between teams. In the following, possible solutions based on the mentioned methods are addressed in more detail, and for further information, refer to Section 3.5.

3.2.1. Secret sharing protocol

SSPs enable developers to share content confidentially with their collaboration DT teams (in this study, commercial information) [10,13]. An SSP works by splitting a secret (such as a cryptographic key) into multiple shares that are distributed to different parties. Each share alone does not provide any information about the secret, but when a sufficient number of shares are combined, the secret can be reconstructed. For DT projects, SSPs can be utilized to:

1. *Enhance Security*: By distributing parts of the commercial information among team members, no single entity holds the complete information, reducing the risk of a full data breach.
2. *Flexible Collaboration*: Team leaders can manage how and when different parts of the commercial information are shared, enabling secure and controlled access to sensitive data.
3. *Dynamic Access Control*: SSPs can be integrated with role-based access control (RBAC) systems to dynamically adjust access permissions based on team roles and project phases [94,95].

3.2.2. Applying threshold schemes

Threshold schemes allow multiple entities to share DT projects' commercial information, provided a minimum number of them (the threshold) are present to reconstruct the data [14]. This scheme is particularly useful in scenarios where high availability and fault tolerance are required. Key benefits of threshold schemes in DT projects include:

1. **Robust Fault Tolerance:** Even if some team members are unavailable or compromised, the commercial information can still be reconstructed and utilized, ensuring project continuity.
2. **Secure MPC:** Threshold schemes can facilitate secure MPC, where computations on commercial data are performed jointly by multiple parties without revealing their individual inputs [14].
3. **Enhanced Privacy:** Combining threshold schemes with other cryptographic tools, such as digital signatures and encryption algorithms, ensures that only authorized sets of parties can access and verify the commercial information, providing an additional layer of privacy and security.

3.2.3. Homomorphic calculation

Homomorphic encryption allows computations to be performed on encrypted data without decrypting it, thus ensuring data privacy during processing [63]. This capability is especially useful for DT projects where distributed teams need to collaborate on sensitive data. Applications of homomorphic encryption in DT projects include:

1. **Confidential Data Analysis:** Teams can perform analytics on encrypted commercial data, obtaining results that are also encrypted, which are then decrypted by authorized entities [96,97].
2. **Secure Data Aggregation:** Project managers can aggregate encrypted contributions from various team members without exposing the individual contributions, preserving confidentiality.
3. **Regulatory Compliance:** Homomorphic encryption helps in complying with data protection regulations (e.g., GDPR) by ensuring that sensitive data is never exposed in plaintext during processing.

3.2.4. Proof-of-Existence (PoE) and Zero-knowledge Proof

Proof-of-Existence (PoE) is an attractive characteristic that is provided by blockchain technology [98–100], and ZKP schemes have received much assistance to it [10]. PoE indicates that a document was created at a specific time without revealing detailed information around the production. As everyone knows, recorded transactions in the blockchain are tamper-proof, and it is impossible to change. Thus, the PoE issuer easily proves the existence of each document at a specific time. Therefore, PoE and ZKP could provide provable ownership and confidentiality so that it is computationally impossible for curious users to obtain the product details. As a result, by applying the mentioned approaches, inventors and producers of a DT demonstrate their ownership rights without revealing actual patent documents and confidential commercial information. Advantages of PoE and ZKPs in DT projects include:

1. **Tamper-Proof Records:** Blockchain's immutable ledger ensures that once a PoE is recorded, it cannot be altered, providing a reliable timestamp and proof of the document's existence.
2. **Confidential Verification:** ZKPs allow one party to prove to another that they know a value without disclosing any information apart from the fact that they know the value. This is useful for verifying ownership or authenticity of commercial information without exposing the actual data.
3. **Secure Intellectual Property Management:** By leveraging PoE and ZKPs, DT developers can securely demonstrate their ownership rights and protect their intellectual property [7], ensuring that sensitive commercial information remains confidential while proving the legitimacy of their claims.

3.3. Fundraising

Experts cooperating with twin teams should be financially supported; therefore, DT teams should have strong financial backgrounds. The need for government or academic funds is required for the growth of twin teams. In fact, funding for DT projects can be obtained via a single source or through multiple investors (in this regard, the *FunDeSci* platform is a leading team provides NFT-based fundraising for early-career

researchers⁶). NFTs linked to DT projects are considered a new application for NFTs and a new reliable method that assists in fundraising for DT projects. For example, as with football clubs' fan tokens (which are known as popular types of tokens), these NFTs are assumed to be twin companies' supporting tokens that guarantee financial funds such as crowdfunding, distributed investing, and sponsoring by huge commercial companies.

Given the high costs associated with the development and maintenance of DTs, diversified funding strategies are crucial. Beyond traditional funding methods, leveraging modern financial technologies and decentralized systems can provide robust alternatives. This section explores various innovative solutions for fundraising in DT projects.

3.3.1. Blockchain-based crowdfunding

Blockchain-based crowdfunding is a well-known issue for funding through smart contracts. It is possible to consider this reliable method for DT projects fundraising to assist DT teams [74]. By utilizing blockchain technology, fundraising can be conducted transparently and securely. Blockchain provides immutable records of transactions, which enhances trust among investors.

Recent developments in Initial Coin Offerings (ICOs) and Security Token Offerings (STOs) present promising opportunities for DT projects (it is highlighted that the *FunDeSci* platform is one of the best ones, in this regard). ICOs allow projects to raise capital by issuing tokens that represent future utility or access to the project's services. Conversely, STOs issue tokens that represent ownership or equity in the project, providing more regulated and secure investment opportunities.

One successful example is the case of XYZ DTs⁷, which raised over 10 Million\$ through a blockchain-based STO. This approach attracted a diverse group of investors and enabled the company to scale rapidly.

3.3.2. Decentralized science (DeSci)

Tokenizing everything relying on blockchain technology makes them reliable and increases trust; the definition of decentralized finance (DeFi) was proposed in 2017 (it means tokenizing financial correspondences and all related issues based on blockchain). As a subgroup of DeFi, decentralized science (DeSci) assists researchers and academia in tokenizing their achievements on a blockchain [74,101,102]. As a result, releasing DT projects as DeSci tokens is an excellent and reliable idea for fundraising in this field. Besides, generated sciences on the Metaverse (e.g., Metaverse-based courses, Metaverse-based universities, Metaverse-based colleges, Metaverse-based scientific presentations, Metaverse-based collaborating achievements, etc.) should be kept secure via DeSci tokens.

The DeSci model promotes a new paradigm where scientific research and technological innovations are funded and governed in a decentralized manner [101,102]. By tokenizing scientific outputs, researchers can secure funding directly from a global pool of investors and enthusiasts. This approach not only democratizes funding but also incentivizes innovation and collaboration across borders.

For instance, the *FunDeSci* team, as a powerful research group, successfully launched a DeSci initiative where each research milestone was tokenized and offered to investors. This model not only funded their projects but also engaged a community of supporters who had a vested interest in the success of the research.

3.3.3. NFTs and DTs

NFTs represent unique digital assets and have revolutionized various industries, including digital art, gaming [65], and now DTs [57]. By linking NFTs to DT projects, companies can create unique, verifiable digital assets that represent parts of the DT or rights to access certain functionalities or data. This method provides an innovative way to raise funds and engage with a broader audience.

⁶ <https://fundesci.com/>.

⁷ <https://digitaltwinstudios.xyz/>.

For example, Khalifa University researchers offered NFTs representing ownership shares in their DT of a smart city⁸. Investors could purchase these NFTs, gaining partial ownership and access to exclusive data generated by the DT. This strategy not only provided substantial initial funding but also created a new revenue stream as the NFTs could be traded in secondary markets.

3.3.4. Government and Academic Grants

While innovative funding mechanisms are critical, traditional funding sources such as government and academic grants remain vital. Governments worldwide recognize the potential of DTs in improving urban planning, healthcare, and various industries [103–105]. Consequently, many countries have established grant programs specifically for DT projects.

For example, the European Union's Horizon 2020 program has allocated significant funds to projects involving DT technology. These grants support research and development, fostering collaboration between academia and industry to advance DT applications.

Academic institutions also play a crucial role in funding and supporting DT research. Universities often have internal grant programs and partnerships with industry to fund cutting-edge research. Collaborations between universities and private companies can leverage academic expertise and industry resources to drive innovation in DTs.

3.3.5. Corporate Sponsorships and Partnerships

Large corporations often seek to invest in emerging technologies to stay ahead of the curve. DT projects offer attractive opportunities for corporate sponsorships and partnerships. Companies can provide funding, resources, and expertise in exchange for early access to innovative technologies and insights.

For instance, McKinsey Corporation partnered with a leading DT startup to develop a digital replica of their manufacturing process⁹. This collaboration not only provided the startup with the necessary funding but also enabled the McKinsey Corporation to optimize their operations using cutting-edge DT technology.

3.3.6. Venture Capital and Private Equity

Venture capital (VC) and private equity (PE) firms are increasingly interested in DT technologies due to their transformative potential [106,107]. These firms provide substantial funding in exchange for equity, enabling DT projects to scale rapidly.

JKL Ventures, a prominent VC firm, recently invested 15 Million\$ in a DT startup focused on healthcare applications. This investment facilitated the development of a comprehensive DT platform for personalized medicine, showcasing the significant impact of VC funding on DT projects.

In conclusion, funding DT projects requires a multifaceted approach, combining traditional and innovative funding mechanisms. Blockchain-based crowdfunding, DeSci, NFTs, government and academic grants, corporate sponsorships, and venture capital are all viable options [65,74,101,102]. By leveraging these diverse funding sources, DT projects can secure the necessary financial support to drive innovation and achieve their goals.

3.4. Distributed collaboration platforms

Because of the complexity of DTs designing, experts and scientists from different domains, such as manufacturing experts, computer engineers, and product managers, have to create DTs cooperatively. Therefore, no individual is able to know every detail of DTs modeling and development. It is also possible that some cooperating companies work

only on a particular part of DT. This issue requires an accessible, reliable, and secure platform for twin teams' data sharing or a shared workspace (physical or virtual).

3.4.1. DAO

Blockchain, a well-known industry technology, supports accessibility and reliability and is a tamper-proof and transparent database [11]. Therefore, it should be applied as a shared and secure platform or common digital space for twin teams. In this regard, DAO [12] provides a virtual workspace where no central authority exists, and twin teams commonly manage the DAO. DAOs, platforms that can involve parallel intelligence, computational experiments, and parallel execution methods [64], are suitable for twin teams. Moreover, DAOs are a practical approach for solving management and control problems for complex systems.

As a result, to establish a secure shared workplace, twin teams have to connect their stakes in DT projects to their shares of DAO control [75]. Regarding safe data sharing, autonomous and distributed management in DAOs, scattered twin teams should share their DT projects on DAO networks.

3.4.2. Manufacturing service collaboration process

Applying several technologies to a product and requests to employ experts from all over the world have forced manufacturers and product managers (in this study, DTs are discussed as products) to internationally collaborating. Additionally, other services and requirements, such as supply chain management, outsourcing productions, marketing, production, and salary payments, are other reasons in this regard [67]. Therefore, the manufacturing service collaboration process for connecting international DT teams and those in sanctioned countries working on a project is a managing challenge.

Metaverse-based collaboration is a newly-emerged platform/paradigm for long-distance connections. Relying on Web3, researchers and developers in sanctioned countries are able to meet, discuss, and cooperate in a common virtual space; Moreover, blockchain technology guarantees their documents and codes immutability. It is noted that most services in the real world (e.g., Metaverse-based marketing, Metaverse-based payment, Metaverse-based supply chain management, Metaverse-based enrolment, Metaverse for academia, etc.) are compatible and applicable in metaverses.

3.4.3. Common source for universal designing

The worldwide DT teams that are far from others and working on a joint DT project are required to have a common source for universal designing [68]. Several solutions exist, such as cloud-based databases, Web 2.0-based workspaces, and virtual private network (VPN)-based remote cooperating. All of the currently-in-used solutions are supported by a centralized infrastructure; however, they all depend on the proper working of their infrastructures, which is a significant weakness.

Regarding the blockchain definition and its goal(s), it is considered a solution for this challenge. It can be applied as an immutable forever and worldwide accessible database for DT teams. Moreover, each new piece of information (new block) is recorded on this database if the majority of users verify its validity. Based on DT project owner(s) policies (e.g., confidentiality, public accessibility limitation, DT teams authentication, project global time, better scalability, etc.), this study believes that using private blockchain is a more suitable solution.

3.4.4. Collaboration on digital platforms

The increasing COVID-19 pandemic in 2019 – 2021 and the worldwide cooperation of industry, academia, businesses, researchers, and developers have caused to increase in requests for remote working and the popularity of collaboration on digital platforms [69]. DAO and Metaverse could be considered new types of digital platforms for remote collaboration. Moreover, numerous architecture and security protocols have been proposed to provide popular features for DT project owners

⁸ <https://khazna.ku.ac.ae/en/publications/nfts-in-smart-cities-vision-applications-and-challenges>.

⁹ <https://www.mckinsey.com/industries/industrials-and-electronics/our-insights/digital-twins-the-key-to-smart-product-development>.

Table 2
The queries and parameters use in the game-based heuristic method security analysis

Query	Description	Notation	Description
Receive	$\mathcal{A}$ can eavesdrop the public channel or receives data sent through public channel	Set up	Determine $\mathcal{A}$'s assumptions
Reveal	$\mathcal{A}$ can steal the server database or smart card	Challenge	The process the $\mathcal{A}$ runs
Send	$\mathcal{A}$ can send data to server or user	Guess	Check calculated challenge is valid or not
		Challenger	The $\mathcal{A}$'s machine calculates the functions

and teams. This study predicts DT project owners applying these platforms to gather DT teammates worldwide in a shared space.

3.4.5. Sharding techniques

Sharding is a scaling technique that divides a system and related duties to all users into several parts (named shards); Each shard then processes its closer found duty [70,71] (it is highlighted that the implementation of sharding should be made so that no change senses in the application layer or user's side). This method is suggested to apply the sharding technique in distributed collaboration platforms, especially worldwide DT teams.

3.5. Security and privacy

Security and privacy are optional user-demanded features, and developers should design their products to provide common security and privacy features. This study believes that cryptographic tools and primitives, based on mathematics, are able to solve DTs' security and privacy challenges [7,10,40,72,76]. The examples below clarify how cryptographic tools solve security and privacy challenges related to DTs.

3.5.1. Confidentiality

Encryption algorithms ensure the confidentiality of DT details as per the demands of twin teams. In blockchain-based systems lacking a central authority, such as a Public Key Infrastructure (PKI) [96] or a certificate issuer, Identity-based Encryption (IBE) [97] is highly recommended for DTs. IBE eliminates the need for pre-distributed keys and allows for secure communications based on user identities, thus simplifying key management and enhancing security. Advanced encryption standards (AES) and homomorphic encryption can also be applied to protect data at rest and during processing, ensuring confidentiality while enabling secure computations on encrypted data without decryption.

3.5.2. Trust on a third party

To ensure the integrity of DT details, self-judgment data authentication protocols are crucial. These protocols empower twin teams to independently detect intrusions, without relying on a central authority or judge, when changes are detected in DT details. Techniques such as Merkle trees and hash chains can be used to create tamper-evident logs, ensuring data integrity and non-repudiation. Additionally, blockchain technology can provide decentralized trust, enabling secure and transparent data sharing among DT stakeholders.

3.5.3. Zero-knowledge proofs

ZKPs are powerful tools for linking the birth certificate of a DT to a NFT. ZKPs allow users to prove DT ownership associated with NFTs without revealing the DT's details or commercial information. This is particularly useful for DT owners who need to maintain confidentiality while demonstrating ownership. Implementing Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (zk-SNARKs) schemes is recommended, as they provide succinct proofs suitable for large and dynamic datasets typical in DT projects. ZKPs can also enhance privacy in data sharing and verification processes, ensuring that sensitive information remains confidential while still being verifiable.

3.5.4. Secret sharing protocols

SSPs divide a secret into n parts so that all n parts should cooperate to reconstruct the secret [10]. Let n the number of twin teams working on a DT project. Therefore, SSPs enable twin teams to share DT project ownership and licenses and confidentiality against outsiders [13].

3.5.5. Threshold schemes

Threshold schemes, including threshold encryption/decryption, digital signatures, and threshold SSPs, allow a process to proceed only if a minimum threshold t out of n parts of a secret are authenticated [14]. For DT projects, particularly those managed by distributed or DAO-based teams, identity-based threshold schemes offer significant advantages. These schemes enable flexible and secure decision-making processes, ensuring that critical operations can only be executed with consensus among a specified number of team members. This approach enhances security, prevents unilateral actions, and fosters collaborative governance in DT project management.

3.5.6. How can analyze security?

After implementing security and privacy preliminaries in DT projects, they should be analysed to ensure they work soundly. In the following, five important security analysis methods are addressed (it is highlighted that in each category several models and types of analysis exist; in this study, we identify one for each).

1. **Heuristic (game-based):** Many proposed anonymous authentication protocols have been analysed through ad hoc methods, but some drawbacks did not discovered. $\mathcal{A}$'s capabilities and threat models are classified in heuristic analysis. In this case, the adversary not only are capable of eavesdrop the channel but also reveals secret data via data recovery through smart card power analysis attack [108]. According to the protocol's attributes, heuristic analysis method has three steps [108,109]. Experiment function, the success function and the probability function. In experiment function, $\mathcal{A}$ performs the process to gain his required information. The success function, specifies how successful is $\mathcal{A}$ in gaining important data. Finally, the possibility function is $\mathcal{A}$'s probability of success in the recovery of secret values. If the probability of success is negligible (ϵ), the latter protocol is secure against assumed $\mathcal{A}$ [109] (see $Succ_{\mathcal{A}}^{Protocol} = Pr[EX P_{\mathcal{A}}^{H.P.}] \leq \epsilon$). Now, we define heuristic method. First, we describe *Reveal*, that empowers $\mathcal{A}$ to guess and find unknown parameters. *Receive* and *Send* are the queries that mean $\mathcal{A}$ can send or receive data to server or users. On the other hand, it can block user flows and *Sends* its dummy flow to server. *Corrupt* is the last query we define. It means $\mathcal{A}$ can corrupt the smart card or server database and extracts stored data [110,111]. These definitions are briefly written in Table 2. All in all, $\mathcal{A}$ is able to send queries including *Reveal*, *Receive* and *Send* data or can *Corrupt* the parameters stored in the user memory or server database. Since all security and privacy concerns are not clarified through heuristic ways, in this section, we mention some $\mathcal{A}$ abilities to analyse in heuristic method in the following:
2. **Formal model (random oracle (RO)):** According to previous definitions and as illustrated in Fig. 6, Probabilistic Polynomial-Time (PPT) $\mathcal{A}$, who has access to the random oracle, engages in a game with Challenger [112]. The game mentioned above is structured according to the phases listed below:

Table 3
The BAN logic rules

Rule	Notation	Description
R_1 : Nonce verification rule (NVR)	$\frac{P \equiv \#(X), P \equiv Q \sim X}{P \equiv Q \equiv X}$	If P believes that X is fresh and P believes that Q sends X . Then, P believes that Q believes X .
R_2 : Freshness rule (FR)	$\frac{P \equiv (X)}{P \equiv (X, Y)}$	If P believes X . Then P believes that both X and Y are fresh.
R_3 : Seeing	$\frac{P \triangleleft (X, Y)}{P \triangleleft X}$	If P can see set (X, Y) . Then P can see the message X .
R_4 : Message meaning rule	$\frac{P \triangleleft X, P \triangleleft (X)_K}{P \equiv Q \sim X}$	If P believes that K is a shared key between P and Q and P received message X which encrypted by K . Then P believes Q sends the message X .
R_5 : Belief(1)	$\frac{P \equiv Q \equiv (X, Y)}{P \equiv Q \equiv X}$	If P believes that Q believes set (X, Y) . Then, P believes Q believes message X .
R_6 : Belief(2)	$\frac{P \equiv Q \sim (X, Y)}{P \equiv Q \sim X}$	If P believes that Q sends set (X, Y) . Then P believes Q sends the message X .

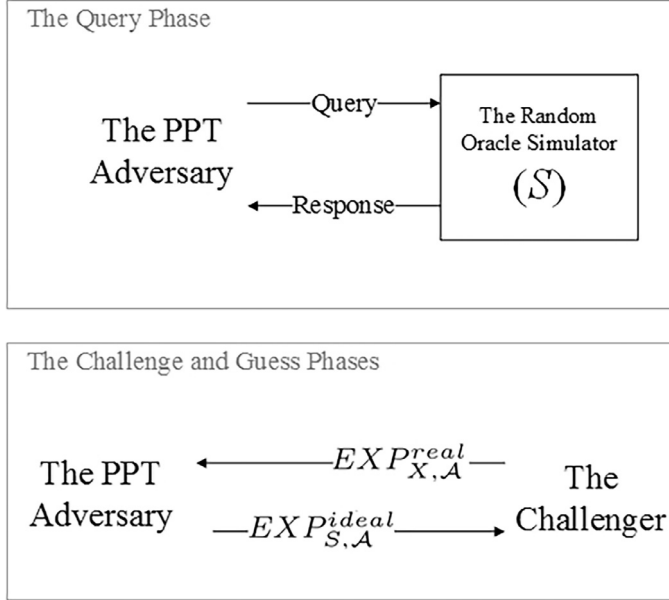


Fig. 6. The RO Model

- **Query:** The $\mathcal{A}$ sends polynomially bounded numbers of queries to the random oracle. The random oracle returns the response related to each sent query to the $\mathcal{A}$ as a real experiment output. The $\mathcal{A}$ collects all responses and stores the query-response table $\{Query, Response\}$ for use in the Guess phase.
 - **Challenge:** The Challenger selects a random value and executes by it the real experiment $EXP_{X,A}^{real}$ and sends the output to $\mathcal{A}$. The Challenger then asks $\mathcal{A}$ to guess an ideal input.
 - **Guess:** The $\mathcal{A}$ has to guess an ideal input for the mentioned experiment in such a way that that the experiment returns the same output that was returned when the experiment had been executed by Challenger.
- The $\mathcal{A}$ wins, and the game returns 1 if $Pr[EXP_{X,A}^{real} = EXP_{S,A}^{ideal}] > \epsilon$; Else, the game returns 0.
3. **Model-based (BAN logic):** Burrows, Abadi, and Needham (BAN) presented a logic model to prove authentication protocols [113] that also is used in nowadays authentication protocols [114]. The notation of the BAN logic and its rules are presented in Tables 3 and 4, respectively. The BAN logic has four steps to prove authentication protocols, including goal, idealization, supposition, and proof.
 4. **Simulation-based (AVISPA):** Automated Validation of Internet Security Protocols and Applications (AVISPA) is a suite of tools designed for the automated analysis and validation of security protocols [115,116]. Its primary goal is to provide a framework that helps in identifying security flaws in protocols, ensuring that they

Table 4
The BAN logic notation

Notation	Description
$P \equiv X$	P believes X
$P \sim X$	P sends message X
$P \triangleleft X$	P receives X
$P \rightarrow Q : M$	P sends M to Q
$P \Rightarrow X$	P has jurisdiction over X
$\#(X)$	X is fresh
$P \leftrightarrow K Q$	K is a shared secret key between P and Q
$P \stackrel{K}{\equiv} X$	The K is a secret formula which, can be used by P and X to prove their identity to another, because only P and X know K
AG_i	i th goal of an adversary
AS_i	i th assumption
M_i	i th message
L_i	i th line of proof
PG_i	i th goal of a protocol
R_i	i th rule
S_i	i th supposition

meet specified security properties. In other words, AVISPA is a comprehensive toolset designed to automate the verification of security protocols. It employs a formal language for protocol specification and integrates multiple back-end tools for thorough analysis. Its inputs are protocol specifications in High-Level Protocol Specification Language (HLPSSL), and its outputs are detailed reports indicating the security status of the protocol, including any found vulnerabilities. The visual interface, particularly the AVISPA Web Interface, aids users in managing and interpreting the protocol analysis process.

- (a) **Environments and Tools:** AVISPA operates within a formal verification environment, utilizing a high-level specification language called HLPSSL. The suite comprises several components, each designed for specific tasks in the protocol verification process:
 - **HLPSSL:** This is the primary language used to model the security protocols. It allows for detailed specification of protocols, including the roles involved, the messages exchanged, and the security properties that need to be verified.
 - **HLPSSL2IF:** A translator that converts HLPSSL specifications into the Intermediate Format (IF), which is a lower-level representation used by the back-end analysis tools.
 - **Back-end Analysis Tools:** AVISPA integrates multiple back-end tools, each employing different techniques to analyze the IF representation of the protocol:
- (b) **Inputs and Outputs:** The primary input to AVISPA tools is the protocol specification written in HLPSSL. This includes defining the roles, the initial state, the transitions (messages exchanged), and the security goals (e.g., confidentiality, authentication). The main output from AVISPA is a detailed analysis report. This includes: (1) Whether the protocol is safe or unsafe with respect to the specified security properties. (2) If unsafe, a trace or coun-

terexample showing how the security property is violated (i.e., an attack scenario).

- (c) *Visual Interface*: AVISPA tools are primarily command-line based, but there is also a graphical user interface called AVISPA Web Interface which simplifies interaction with the toolset. *AVISPA Web Interface*: This provides a more user-friendly way to specify protocols and view results. Users can upload HLPSP files, run analysis tools, and visualize the results through a web-based interface. It helps in managing the different stages of protocol analysis, from specification to result interpretation.
 - (d) *Example Workflow*: Generally, the AVISPA tool executes in the four below steps:
 - (1) *Specification*: The user writes a protocol specification in HLPSP.
 - (2) *Translation*: The HLPSP specification is translated to IF using HLPSP2IF.
 - (3) *Analysis*: The IF representation is analyzed using one or more of the back-end tools (OFMC, CL-AtSe, SATMC, TA4SP).
 - (4) *Interpretation*: The results are interpreted to determine if the protocol meets the security requirements. If not, the provided counterexamples are used to understand and fix the issues.
5. **Advanced simulation-based (TAMARIN prover)**: The TAMARIN Prover is a sophisticated tool used for the formal analysis and verification of security protocols [117,118]. It allows researchers and developers to model and analyze the security properties of various protocols to ensure their robustness against potential attacks. Here's an overview of its main components and functionalities:
- *Environments*: TAMARIN can be installed on Unix-like operating systems, including Linux and macOS. It can also be run on Windows using virtualization tools or Windows Subsystem for Linux (WSL). It requires certain dependencies like OCaml, Haskell, and other libraries, which can be installed via package managers. TAMARIN has an interactive mode where users can incrementally build and check their models. This is particularly useful for complex protocols and iterative development.
 - *Tools and Functionalities*: TAMARIN uses a custom specification language to model security protocols. This language allows users to define messages, actions, and rules that represent the behavior of the protocol. Protocols are modeled using multiset rewriting rules, which describe how the state of the protocol changes over time. Users can specify security properties using a temporal logic known as TAMARIN's built-in logic. These properties can include confidentiality, integrity, authentication, and more. TAMARIN supports both trace properties (which consider individual protocol executions) and equivalence properties (which consider the indistinguishability of different protocol executions). TAMARIN provides automated reasoning capabilities to prove or disprove the specified properties. For more complex proofs, TAMARIN offers interactive proof capabilities where users can guide the proof search process.
 - *Inputs and Outputs*: Written in TAMARIN's specification language, detailing the protocol steps, messages exchanged, and possible actions. Defined using TAMARIN's temporal logic, specifying what security guarantees need to be verified. TAMARIN provides detailed outputs on whether the specified security properties hold. It includes counterexamples if a property is violated, helping users understand the failure scenarios. The tool generates proof scripts that can be examined and modified for further analysis. Detailed logs and reports on the proof process, including steps taken and reasoning paths explored.
 - *Visual Interface*: TAMARIN offers a web-based graphical user interface that facilitates the interaction with the tool. It provides visualization of the protocol states, actions, and proof steps. Users can navigate through the model, examine proof steps, and manage their analysis in an intuitive manner. The GUI includes visualization tools that help users understand the protocol's execu-

tion flow and the structure of the proofs. This can be especially helpful for debugging and refining the protocol models.

The TAMARIN Prover is a powerful tool for the formal analysis of security protocols, providing a rich environment for modelling, analyzing, and verifying security properties. With its combination of automated and interactive proofs, detailed outputs, and a user-friendly graphical interface, TAMARIN facilitates rigorous security analysis, aiding in the development of robust and secure communication protocols.

4. Discussion

As discussed in the previous section, connecting DTs to NFTs commits various benefits for DT projects and solves IT-based DT challenges. Therefore, DTs are stored on a blockchain as a public ledger, a publicly shared database for DT teams, and no one can change or tamper with them. Additionally, Blockchain allows monetary value to be assigned to DTs of physical assets, intellectual properties, and patents under the distributed ledger technology and prevents the risk of duplications. As a result, NFTs are paving the way for a new direction of trust in the digital world and are the future trend. In addition, DTs will get benefits from NFTs. As a brief conclusion of the previous section, we provide a SWOT (Strongness - Weakness - Opportunity - Threat) analysis in Table 5. Table 5 strongly clarifies the application of the aforementioned technologies to apply in DT projects. It is highlighted that, Table 5 provides a general analysis, and the specific strengths, weaknesses, opportunities, and threats may vary depending on the specific application and implementation of these technologies in a DT project.

To strike a balance between privacy protection and regulation, developing NFT-based DTs on consortium blockchains is a suggested alternative since it restricts public access, and only qualified users/devices can send/receive data [11]. This feature solves the concerns of twin teams about DTs' commercial information and data sharing. Despite the accessibility of qualified users/devices who have been authenticated and who should follow data formatting, unstructured data and unfavourable flows do not appear in the consortium blockchain. Additionally, as a shared database, consortium blockchain provides the properties of public blockchain for authenticated users, such as accessibility, transparency, and immutability. Therefore, twin teams have an accessible, reliable, secure database for DT data sharing and a shared workspace.

Incorporating advanced cryptographic primitives such as ZKPs and homomorphic encryption further enhances the privacy and security of DT data. ZKPs enable the verification of data without revealing the data itself, ensuring that sensitive information remains confidential while still allowing for robust validation processes. Homomorphic encryption allows computations to be performed on encrypted data, providing a layer of privacy crucial for sensitive industrial applications where data confidentiality is paramount. Moreover, integrating DeSci principles can democratize the research and development process within DT ecosystems. By leveraging blockchain technology for transparent and decentralized funding mechanisms, innovation can be accelerated through broader participation from the global scientific community. This can lead to more diverse and inclusive development processes, fostering collaboration across various disciplines and geographies.

Distributed crowdfunding models facilitated by blockchain can also revolutionize the financing of DT projects. These models provide a transparent and efficient way to raise funds, allowing stakeholders from around the world to contribute to and invest in DT innovations. This approach not only democratizes the funding process but also builds a community of engaged supporters who can contribute valuable insights and resources.

Autonomous management systems, powered by smart contracts, can further enhance the efficiency and reliability of DT operations. These systems can automate routine tasks, enforce predefined rules, and ensure compliance with regulatory standards without human intervention.

Table 5

The SWOT analysis of the application of the aforementioned technologies to apply in DT projects

	Strongness	Weakness	Opportunity	Threat
Blockchain technology as infrastructure	Inheriting blockchain exciting features.	Immutability can be assumed as a weakness for fully blockchain-based ideas.	Existing several innovative blockchains and powerful blockchain developer teams all around the world.	Wrong blockchain selection, and the 51% attack.
Using NFTs	Uniqueness, provided by NFTs, is the best feature required for novel DT projects.	Everyone does not accept. Being new causes several weaknesses.	Trending and state-of-the-art technology, and the possibility of high growth in the short term.	Multiple ownership is challenging for designers and developers which can be made cyber frauds.
DAO-based DT teams	Relying on blockchain technology empowered by cryptographic algorithms.	As some of DT projects' details are private and sensitive, public DAOs do not support data confidentiality.	Unlimited collaboration with worldwide DT teams.	Lack of attention to DAO developers and wrong DAO selection cause using premature paradigms and increase cyberattacks.
Metaverse-based common workspaces	Providing natural environments as good as physical common workspace.	Fully Metaverse-based workspaces require powerful processors and broadband Internet connections (expensive). It seems that there is a need for launching a specialized metaverse.	There are many Metaverse-related tools and UIs which facilitate DT teams working.	Weaknesses of security protocols (e.g., authentication, access control, etc.) are threats to DT data integrity.
Edge computing	Main features of edges (e.g., low latency) can provide many improvements in distributed DT projects.	Need for equipment for edges so that also be compatible with blockchain-based networks.	Good compatibility of edge devices with synchronized and Internet-based projects.	Off-chain computation increases the possibility of fraud in blockchain-based projects.
Parallel computing	Enhanced processing speed. Efficient handling of large datasets. Improved real-time analytics.	High infrastructure costs. Complexity in programming and management	Better simulation and modeling. Potential for breakthroughs in real-time applications	Security vulnerabilities. High energy consumption
Using SSPs	Enhanced data security. Robustness against single point of failure.	Increased computational overhead. Complexity in implementation	Secure multi-party computation. Improved confidentiality in collaborations.	Risk of collusion and management complexity
Applying threshold schemes	Improved security and fault tolerance. Decentralized control.	Increased complexity. Potential performance overhead.	Enhanced collaborative security. More resilient system designs.	Risk of partial collusion. Implementation challenges.
Homomorphic calculation	Ability to perform computations on encrypted data. Enhanced privacy.	High computational cost. Limited by current technology efficiency.	Privacy-preserving data analytics. Secure cloud computing.	Vulnerability to specific cryptographic attacks. Limited practical implementations.
PoE mechanism Using ZKP	Strong privacy guarantees. Reduced need for trust in verifiers.	Computationally intensive. Complexity in development and verification.	Secure authentication systems. Enhanced privacy in transactions.	Potential vulnerabilities in protocol implementation. Scalability issues.
DeSci-based projects	Democratization of research. Increased transparency and collaboration.	Regulatory uncertainties. Resistance from traditional institutions	Innovative funding models. Broader participation in research.	Intellectual property concerns. Potential for low-quality research
Distributed crowdfunding	Increased access to funding. Enhanced transparency and trust.	Risk of fraud and scams. Regulatory challenges.	Democratized investment opportunities. Enhanced community engagement.	Regulatory crackdowns. Market saturation
Distributed ledger as a database	High transparency. Tamper-proof records.	Scalability issues. High storage requirements	Improved auditability. Enhanced data integrity.	Regulatory and compliance issues. Technological adoption barriers.
Autonomous management	Reduced need for human intervention. Enhanced efficiency and scalability.	Potential lack of accountability. Ethical and legal concerns.	Streamlined operations. Innovative business models.	Malfunctioning or rogue algorithms. Security vulnerabilities
Worldwide collaboration	Diverse perspectives and expertise. Enhanced innovation and problem-solving.	Coordination challenges. Cultural and language barriers.	Global partnerships. Broader talent pool.	Geopolitical risks. Intellectual property disputes.

However, the implementation of such systems must be carefully managed to mitigate risks associated with algorithmic biases and operational errors.

The worldwide collaboration enabled by blockchain technology can transform how DT projects are managed and developed. By providing a decentralized and secure platform for data sharing and communication, blockchain facilitates seamless collaboration among stakeholders, regardless of their geographical locations. This global approach can accelerate innovation and improve the resilience of DT systems by incorporating diverse perspectives and expertise.

As DT technology continues to evolve, addressing state-of-the-art issues such as scalability, interoperability, and regulatory compliance will be critical. Ensuring that consortium blockchains can handle the growing volume of data and transactions is essential for maintaining performance and reliability. Interoperability between different blockchain networks and traditional IT systems is necessary to create a cohesive and integrated DT ecosystem. Finally, navigating the complex regulatory landscape requires ongoing dialogue with policymakers and adherence to emerging standards and best practices. By addressing these challenges, DT projects can fully realize their potential and drive significant advancements in various industries.

5. Conclusion

NFTs enhance the assets' ownership reliability, and they provide potent motivation for implementing NFT-based DTs on a blockchain. Additionally, this NFT technology has found its place in many applications such as digital real estate, artworks, vehicles, and games. Nevertheless, birth certificates of DTs, intellectual properties, and patent registrations are considered new applications for NFTs and are critical for scientists, product managers, engineering, and industries. Therefore, generating NFTs of mentioned items and connecting them to the real world allow the transfer of provable ownership of patents, shared workspace for co-operating twin teams, and intellectual properties for DTs in the digital world.

The combination of consortium blockchain and cryptographic techniques could answer all security and privacy challenges in distributed systems. In fact, in a consortium blockchain, implementing policies, including security updates, key management, and revoking misbehaved users/devices are achievable. This study believes that the digital world is the future trend, so DTs will be one of the main connectors between the real and digital worlds since they can provide the most natural sense for users in the digital world [57]. Despite this fact and the ability of NFTs, NFTs will play a critical role in the future of Metaverse and DTs.

Undoubtedly, the digital world will be one of the future trends that is welcomed by industrial owners. As a newly trended, Metaverse has attracted people in various fields of technology, investment, games, market, and especially industry. Regarding the significant role of DTs in the industry, they will play critical roles in the digital world (Metaverse), and connecting DTs to NFTs helps their development in the digital world.

Ethical Approval

Consent to participate is "Not applicable" and Consent for publication is "Not applicable".

Availability of data and materials

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no conflict of interest to the publication of this article.

CRediT authorship contribution statement

Seyed Mojtaba Hosseini Bamakan: Conceptualization, Investigation, Validation, Writing – review & editing. **Saeed Banaeian Far:** Conceptualization, Investigation, Formal analysis, Validation, Writing – review & editing, Supervision.

Acknowledgment

We, as authors, appreciate anonymous reviewers for their insightful comments to improve this work.

References

- [1] F. Tao, Q. Qi, Make more digital twins, *Nature* 573 (2019) 490–491.
- [2] D. Jones, C. Snider, A. Nassehi, J. Yon, B. Hicks, Characterising the digital twin: A systematic literature review, *CIRP J. Manuf. Sci. Technol.* 29 (2020) 36–52.
- [3] F. Zhang, J. Bai, D. Yang, Q. Wang, Digital twin data-driven proactive job-shop scheduling strategy towards asymmetric manufacturing execution decision, *Sci. Rep.* 12 (1) (2022) 1–19.
- [4] S. Banaeian Far, A. Imani Rad, Internet of artificial intelligence (IoAI): the emergence of an autonomous, generative, and fully human-disconnected community, *Discov. Appl. Sci.* 6 (3) (2024) 1–16.
- [5] K. Costello, G. Omale, Gartner survey reveals digital twins are entering mainstream use, 2019, Gartner Inc.) Retrieved April. 30, 2020.
- [6] C. Quilodrán-Casas, V.L. Silva, R. Arcucci, C.E. Heaney, Y. Guo, C.C. Pain, Digital twins based on bidirectional LSTM and GAN for modelling the COVID-19 pandemic, *Neurocomputing* 470 (2022) 11–28.
- [7] S.M.H. Bamakan, N. Nezhadastani, O. Bodaghi, Q. Qu, Patents and intellectual property assets as non-fungible tokens; key technologies and challenges, *Sci. Rep.* 12 (1) (2022) 1–13.
- [8] F. Valeonti, A. Bikakis, M. Terras, C. Speed, A. Hudson-Smith, K. Chalkias, Crypto collectibles, museum funding and openGLAM: challenges, opportunities and the potential of non-fungible tokens (NFTs), *Appl. Sci.* 11 (21) (2021) 9931.
- [9] P. Bauer, P.D. Dueben, T. Hoefler, T. Quintino, T.C. Schulthess, N.P. Wedi, The digital revolution of earth-system science, *Nature Comput. Sci.* 1 (2) (2021) 104–113.
- [10] F. Benhamouda, C. Gentry, S. Gorbunov, S. Halevi, H. Krawczyk, C. Lin, L. Reyzin, Can a public blockchain keep a secret? in: *In Theory of Cryptography Conference*, Springer, Cham, 2020, pp. 260–290.
- [11] C.V. Helliar, L. Crawford, L. Rocca, C. Teodori, M. Veneziani, Permissionless and permissioned blockchain diffusion, *Int. J. Inf. Manag.* 54 (2020) 102136.
- [12] S.B. Far, S.M.H. Bamakan, Blockchain-based reporting protocols as a collective monitoring mechanism in DAOs, *Data Sci. Manag.* 5 (1) (2022) 11–12.
- [13] W. Zheng, K. Wang, F.Y. Wang, Gan-based key secret-sharing scheme in blockchain, *IEEE Trans. Cybernetic.* 51 (1) (2020) 393–404.
- [14] K. Yu, L. Tan, C. Yang, K.K.R. Choo, A.K. Bashir, J.J. Rodrigues, T. Sato, A blockchain-based shamir's threshold cryptography scheme for data protection in industrial internet of things settings, *IEEE IoT J.* (2021).
- [15] P. Raj, Empowering digital twins with blockchain, in: *In Advances in Computers*, volume 121, Elsevier, 2021, pp. 267–283.
- [16] Y. Wang, Z. Su, S. Guo, M. Dai, T.H. Luan, Y. Liu, A survey on digital twins: architecture, enabling technologies, security and privacy, and future prospects, *IEEE IoT J.* (2023).
- [17] H. van der Valk, H. Haße, F. Möller, B. Otto, Archetypes of digital twins, *Bus. Inf. Syst. Eng.* 64 (3) (2022) 375–391.
- [18] R. Minerva, N. Crespi, Digital twins: Properties, software frameworks, and application scenarios, *IT Profession.* 23 (1) (2021) 51–55.
- [19] M. Schluse, M. Priggemeyer, L. Atorf, J. Rossmann, Experimentable digital twin-streamlining simulation-based systems engineering for industry 4.0, *IEEE Trans. Ind. Inform.* 14 (4) (2018) 1722–1731.
- [20] D.J. Wagg, K. Worden, R.J. Barthorpe, P. Gardner, Digital twins: state-of-the-art and future directions for modeling and simulation in engineering dynamics applications, *ASCE-ASME J. Risk Uncert Engrg. Sys. Part B Mech. Engrg.* 6 (3) (2020).
- [21] Y. Jiang, S. Yin, K. Li, H. Luo, O. Kaynak, Industrial applications of digital twins, *Philos. Trans. R. Soc. A* 379 (2207) (2021) 20200360.
- [22] V. Kamath, J. Morgan, M.I. Ali, Industrial iot and digital twins for a smart factory: An open source toolkit for application design and benchmarking, In *2020 Global Internet of Things Summit (GloTS)* (2020) 1–6.
- [23] A. Fuller, Z. Fan, C. Day, C. Barlow, Digital twin: Enabling technologies, challenges and open research, *IEEE access* 8 (2020) 108952–108971.
- [24] M. Shahzad, M.T. Shafiq, D. Douglas, M. Kassem, Digital twins in built environments: An investigation of the characteristics, applications, and challenges, *Buildings* 12 (2) (2022) 120.
- [25] R. Sahal, S.H. Alsamhi, J.G. Breslin, K.N. Brown, M.I. Ali, Digital twins collaboration for automatic erratic operational data detection in industry 4.0, *Appl. Sci.* 11 (7) (2021) 3186.

- [26] B. Björnsson, C. Borrebaeck, N. Elander, T. Gasslander, D.R. Gaweł, M. Gustafsson, M. Benson, Digital twins to personalize medicine, *Genome Med.* 12 (1) (2020) 1–4.
- [27] T. Vats, S.K. Singh, S. Kumar, B.B. Gupta, S.S. Gill, V. Arya, W. Alhalabi, Explainable context-aware iot framework using human digital twin for healthcare, *Multimedia Tools Appl.* (2023) 1–25.
- [28] G.N. Nguyen, N.H. Le Viet, M. Elhoseny, K. Shankar, B.B. Gupta, A.A. Abd El-Latif, Secure blockchain enabled cyber-physical systems in healthcare using deep belief network with resnet model, *J. Parallel Distrib. Comput.* 153 (2021) 150–160.
- [29] A. Raj, S. Prakash, A privacy-preserving authentic healthcare monitoring system using blockchain, *Int. J. Softw. Sci. Comput. Intell. (IJSSCI)* 14 (1) (2022) 1–23.
- [30] T. Vats, S.K. Singh, S. Kumar, B.B. Gupta, S.S. Gill, V. Arya, W. Alhalabi, Explainable context-aware iot framework using human digital twin for healthcare, *Multimedia Tool Appl* (2023) 1–25.
- [31] W.M. Mohammed, R.E. Haber, J.L. Martinez Lastra, Ontology-driven guidelines for architecting digital twins in factory automation applications, *Machines* 10 (10) (2022) 861.
- [32] A. Göppert, L. Grah, J. Rachner, D. Grunert, S. Hort, R.H. Schmitt, Pipeline for ontology-based modeling and automated deployment of digital twins for planning and control of manufacturing systems, *J. Intell. Manuf.* (2021) 1–20.
- [33] Q. Qi, F. Tao, Digital twin and big data towards smart manufacturing and industry 4.0: 360 degree comparison, *Ieee Access* 6 (2018) 3585–3593.
- [34] K.M. Alam, A. El Saddik, C2PS: A digital twin architecture reference model for the cloud-based cyber-physical systems, *IEEE access* 5 (2017) 2050–2062.
- [35] U. Zaman, I. Koo, S. Abbasi, S.H. Raza, M.G. Qureshi, Meet your digital twin in space? profiling international expat's readiness for metaverse space travel, *Tech-Saviness, COVID-19 Travel Anxiety, and Travel Fear of Missing Out. Sustainability* 14 (11) (2022) 6441.
- [36] Y. Qamsane, J.R. Phillips, C. Savaglio, D. Warner, S.C. James, K. Barton, Open process automation-and digital twin-based performance monitoring of a process manufacturing system, *IEEE Access* 10 (2022) 60823–60835.
- [37] M. Eckhart, A. Ekelhart, R. Eisl, Digital twins for cyber-physical threat detection and response, *ERCIM News: Spec. Theme Smart Circular Cities* 127 (2021) 12–13.
- [38] G.N. Nguyen, N.H. Le Viet, M. Elhoseny, K. Shankar, B.B. Gupta, A.A. Abd El-Latif, Secure blockchain enabled cyber-physical systems in healthcare using deep belief network with resnet model, *Journal of parallel and distributed computing* 153 (2021) 150–160.
- [39] A. Sharma, E. Kosasih, J. Zhang, A. Brintrup, A. Calinescu, Digital twins: State of the art theory and practice, challenges, and open research questions, *J. Ind. Inf. Integr.* (2022) 100383.
- [40] K. Alshammari, T. Beach, Y. Rezgui, Cybersecurity for digital twins in the built environment: current research and future directions, *J. Inf. Technol. Construct.* 26 (2021) 159–173.
- [41] D. Holmes, M. Papathanasaki, L. Maglaras, M.A. Ferrag, S. Nepal, H. Janicke, Digital twins and cyber security-solution or challenge?. in: *2021 6th south-east europe design automation, computer engineering, in: Computer Networks and Social Media Conference (SEEDA-CECNSM)*, IEEE, 2021, pp. 1–8.
- [42] S. Nakamoto, Bitcoin: A peer-to-peer electronic cash system, *Decentralized Bus. Rev.* (2008) 21260.
- [43] R. Zhang, R. Xue, L. Liu, Security and privacy on blockchain, *ACM Comput. Surv. (CSUR)* 52 (3) (2019) 1–34.
- [44] S.M.H. Bamakan, A. Motavali, A.B. Bondarti, A survey of blockchain consensus algorithms performance evaluation criteria, *Expert Syst. Appl.* 154 (2020) 113385.
- [45] J. Rohr, A. Wright, Blockchain-based token sales, initial coin offerings, and the democratization of public capital markets, *Hast. LJ* 70 (2018) 463.
- [46] S.B. Far, S.M.H. Bamakan, Q. Qu, Q. Jiang, A review of non-fungible tokens applications in the real-world and metaverse, *Procedia Comput. Sci.* 214 (2022) 755–762.
- [47] A. Mekacher, A. Bracci, M. Nadini, M. Martino, L. Alessandretti, L.M. Aiello, A. Baronchelli, Heterogeneous rarity patterns drive price dynamics in NFT collections, *Sci. Rep.* 12 (1) (2022) 1–9.
- [48] M. Solouki, S.M.H. Bamakan, An in-depth insight at digital ownership through dynamic NFTs, *Procedia Comput. Sci.* 214 (2022) 875–882.
- [49] S. Banaeian Far, M. Rajabzadeh Asaar, A. Haghighi, An unlinkable reputation transfer framework for blockchain-based retail markets using non-fungible tokens, *J. Adv. Comput. Eng. Technol.* (2022).
- [50] S. Suhail, R. Hussain, R. Jurdak, A. Oracevic, K. Salah, C.S. Hong, R. Matulevičius, Blockchain-based digital twins: research trends, issues, and future challenges, *ACM Comput. Surv. (CSUR)* 54 (11s) (2022) 1–34.
- [51] I. Yaqoob, K. Salah, M. Uddin, R. Jayaraman, M. Omar, M. Imran, Blockchain for digital twins: Recent advances and future research challenges, *IEEE Netw.* 34 (5) (2020) 290–298.
- [52] S. Hassan, P. De Filippi, Decentralized autonomous organization, *Int. Policy Rev.* 10 (2) (2021) 1–10.
- [53] N. Diallo, W. Shi, L. Xu, Z. Gao, L. Chen, Y. Lu, G. Turner, egov-DAO: A better government using blockchain based decentralized autonomous organization, in: *2018 International Conference on eDemocracy & eGovernment (ICEDEG)*, IEEE, 2018, pp. 166–171.
- [54] D. Asaoka, A behavioral analysis of whistleblowing at Japanese firms, *Human. Soc. Sci. Commun.* 7 (1) (2020) 1–8.
- [55] T.R. Gadekallu, T. Huynh-The, W. Wang, G. Yenduri, P. Ranaweera, Q.V. Pham, M. Liyanage, Blockchain for the metaverse: A review, 2022, *ArXiv preprint arXiv:2203.09738*.
- [56] S. Mystakidis, Metaverse, *Encyclopedia* 2 (1) (2022) 486–497.
- [57] S.B. Far, A.I. Rad, Applying digital twins in metaverse: User interface, security and privacy challenges, *J. Metaverse* 2 (1) (2022) 8–16.
- [58] M. C., Introduction to the metaverse - newzoo trend report 2021, 2021, (????). Newzoo. Available at <https://newzoo.com/insights/trend-reports/newzoo-intro-to-the-metaverse-report-2021-free-version>.
- [59] P.P. Montaz, Some very simple economics of web3 and the metaverse, Available at SSRN (2022).
- [60] F.Y. Wang, R. Qin, X. Wang, B. Hu, Metasocieties in metaverse: metaeconomics and metamanagement for metaenterprises and metacities, *IEEE Trans. Comput. Soc. Syst.* 9 (1) (2022) 2–7.
- [61] S.B. Far, A.I. Rad, S.M.H. Bamakan, M.R. Asaar, Toward metaverse of everything: Opportunities, challenges, and future directions of the next generation of visual/virtual communications, *J. Netw. Comput. Appl.* (2023) 103675.
- [62] S.B. Far, A.I. Rad, Security analysis of big data on internet of things, 2018, (????). *ArXiv preprint arXiv:1808.09491*.
- [63] C. Gentry, Fully homomorphic encryption using ideal lattices, In *Proceedings of the forty-first annual ACM symposium on Theory of computing* (2009) 169–178.
- [64] W.W. Ding, X. Liang, J. Hou, G. Wang, Y. Yuan, J. Li, F.Y. Wang, Parallel governance for decentralized autonomous organizations enabled by blockchain and smart contracts, In *2021 IEEE 1st International Conference on Digital Twins and Parallel Intelligence (DTPi)* (2021) 1–4.
- [65] S. Banaeian Far, S.M. Hosseini Bamakan, NFT-based identity management in metaverses: challenges and opportunities, *SN Appl. Sci.* 5 (10) (2023) 260.
- [66] S.B. Far, M.R. Asaar, A blockchain-based anonymous reporting system with no central authority: Architecture and protocol, *Cyber Secur. Appl.* 2 (2024) 100032.
- [67] F. Tao, Y. Zhang, Y. Cheng, J. Ren, D. Wang, Q. Qi, P. Li, Digital twin and blockchain enhanced smart manufacturing service collaboration and management, *J. Manuf. Syst.* (2020).
- [68] D. Burnett, J. Thorp, D. Richards, K. Gorkovenko, D. Murray-Rust, Digital twins as a resource for design research, 2019, In *Proceedings of the 8th ACM International Symposium on Pervasive Displays*, 1–2.
- [69] W. Reim, E. Andersson, K. Eckerwall, Enabling collaboration on digital platforms: a study of digital twins, *Int. J. Product. Res.* (2022) 1–17.
- [70] M. Zamani, M. Movahedi, M. Raykova, Rapidchain: Scaling blockchain via full sharding, In *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security* (2018) 931–948.
- [71] G. Wang, Z.J. Shi, M. Nixon, S. Han, Sok: Sharding on blockchain, In *Proceedings of the 1st ACM Conference on Advances in Financial Technologies* (2019) 41–61.
- [72] M. Eckhart, A. Ekelhart, Digital twins for cyber-physical systems security: State of the art and outlook, *Secur. Qual. Cyber-Phys. Syst. Eng.* (2019) 383–412.
- [73] T. Kumar, E. Harjula, M. Ejaz, A. Manzoor, P. Poramange, I. Ahmad, M. Ylianttila, Blockchain: Blockchain-edge framework for industrial iot networks, *IEEE Access* 8 (2020) 154166–154185.
- [74] S.B. Far, A.I. Rad, M.R. Asaar, Blockchain and its derived technologies shape the future generation of digital businesses: a focus on decentralized finance and the metaverse, *Data Sci. Manag.* 6 (3) (2023) 183–197.
- [75] A. Imani Rad, S. Banaeian Far, Socialfi transforms social media: an overview of key technologies, challenges, and opportunities of the future generation of social media, *Soc. Netw. Anal. Min.* 13 (1) (2023) 42.
- [76] A. Raj, S. Prakash, A privacy-preserving authentic healthcare monitoring system using blockchain, *Int. J. Softw. Sci. Comput. Intell. (IJSSCI)* 14 (1) (2022) 1–23.
- [77] J. Xu, S. Liu, W. Yang, M. Fang, Y. Pan, Beyond reality: Exploring user experiences in the metaverse art exhibition platform from an integrated perspective, *Electronics* 13 (6) (2024) 1023.
- [78] S.G.M. Habil, S. El-Deeb, N. El-Bassiouny, The metaverse era: leveraging augmented reality in the creation of novel customer experience, *Manag. Sustain.: An Arab Rev.* 3 (1) (2024) 1–15.
- [79] S. Yang, Storytelling and user experience in the cultural metaverse, *Heliyon* 9 (4) (2023).
- [80] Z. Lv, S. Xie, Artificial intelligence in the digital twins: State of the art, challenges, and future research topics, *Digital Twin* 1 (2022) 12.
- [81] P. Unal, O. Albayrak, M. Jomaa, A.J. Berre, Data-driven artificial intelligence and predictive analytics for the maintenance of industrial machinery with hybrid and cognitive digital twins, in: *In Technologies and Applications for Big Data Value*, Springer International Publishing, Cham, 2022, pp. 299–319.
- [82] P. Radanliev, D. De Roure, R. Nicolescu, M. Huth, O. Santos, Digital twins: Artificial intelligence and the iot cyber-physical systems in industry 4.0, *Int. J. Intell. Robot. Appl.* 6 (1) (2022) 171–185.
- [83] K. de Koning, J. Broekhuijsen, I. Kühn, O. Ovaskainen, F. Taubert, D. Endresen, V. Grimm, Digital twins: dynamic model-data fusion for ecology, *Trend. Ecol. Evol.* (2023).
- [84] W. Zhao, C. Zhang, J. Wang, S. Wang, D. Lv, F. Qin, Research on digital twin driven rolling bearing model-data fusion life prediction method, *Ieee Access* (2023).
- [85] F.P. Knebel, R. Trevisan, G.S. do Nascimento, M. Abel, J.A. Wickboldt, A study on cloud and edge computing for the implementation of digital twins in the oil & gas industries, *Comput. Ind. Eng.* 182 (2023) 109363.
- [86] E. Bozkaya, Digital twin-assisted and mobility-aware service migration in mobile edge computing, *Comput. Netw.* 231 (2023) 109798.
- [87] I. Abdullahi, S. Longo, M. Samie, Towards a distributed digital twin framework for predictive maintenance in industrial internet of things (IIoT), *Sensors* 24 (8) (2024) 2663.
- [88] D. Karnehm, A. Samanta, A. Neve, S. Williamson, Five-layer IoT and Fog Computing Framework Towards Digital Twinning of Battery Management Systems for e-Transportation, *IEEE*, 2024, p. 1–7.
- [89] F. Viola, G. Del Corso, R. De Paulis, R. Verzicco, GPU accelerated digital twins of the human heart open new routes for cardiovascular research, *Sci. Rep.* 13 (1) (2023) 8230.

- [90] S. Somanath, V. Naserentin, O. Eleftheriou, D. Sjölie, B.S. Wästberg, A. Logg, Towards urban digital twins: A workflow for procedural visualization using geospatial data, *Remote Sens.* 16 (11) (2024) 1939.
- [91] S. Abdul Rahman, S. Otoum, O. Bouachir, A. Mourad, Management of digital twin-driven iot using federated learning, *IEEE J. Sel. Areas Commun.* (2023).
- [92] L. Jiang, Y. Liu, H. Tian, L. Tang, S. Xie, Resource efficient federated learning and DAG blockchain with sharding in digital twin driven industrial iot, *IEEE IoT J.* (2024).
- [93] M.M. Salim, A. El Azaoui, X. Deng, J.H. Park, FL-CTIF: A federated learning based CTI framework based on information fusion for secure IIot, *Inf. Fusion* 102 (2024) 102074.
- [94] Y.A. Marquis, From theory to practice: Implementing effective role-based access control strategies to mitigate insider risks in diverse organizational contexts, *J. Eng. Res. Rep.* 26 (5) (2024) 138–154.
- [95] B. Yang, H. Hu, Resiliency analysis of role-based access control via constraint enforcement and mathematical programming, *IEEE Trans. Syst. Man, Cybernetic.: Syst.* (2024).
- [96] U. Maurer, Modelling a public-key infrastructure, in: *In Computer Security-E-SORICS 96: 4th European Symposium on Research in Computer Security Rome*, volume 4, Springer Berlin Heidelberg, Italy, 1996, pp. 325–350. September 25–27, 1996 Proceedings
- [97] D. Boneh, M. Franklin, Identity-based encryption from the weil pairing, in *Annual International Cryptology Conference* (2001) 213–229.
- [98] K. Chopra, K. Gupta, A. Lambora, Proof of existence using blockchain, in: *In 2019 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COMITCon)*, IEEE, 2019, pp. 429–431.
- [99] L.W.M. Shawn, P. Murali Mohan, P. Loh Kok Keong, V. Balachandran, Blockchain-based proof of existence (poe) framework using ethereum smart contracts, in: *In Proceedings of the Eleventh ACM Conference on Data and Application Security and Privacy*, 2021, pp. 301–303.
- [100] K. Ravi Kishore, G. Jyostna, M.U. Patil, P.R. Lakshmi Eswari, E. Magesh, Blockchain based proof of existence (poe) application for educational certificate verification, *In Communication Software and Networks: Proceedings of INDIA 2019* (2021) 575–586.
- [101] W. Ding, J. Hou, J. Li, C. Guo, J. Qin, R. Kozma, F.Y. Wang, Desci based on web3 and DAO: A comprehensive overview and reference model, *IEEE Trans. Computat. Soc. Syst.* 9 (5) (2022) 1563–1573.
- [102] W. Ding, J. Li, R. Qin, R. Kozma, F.Y. Wang, A new architecture and mechanism for decentralized science metamarkets, *IEEE Trans. Syst. Man Cybernetic.: Syst.* (2023).
- [103] M.D. Muhlheim, P. Ramuhalli, A. Huning, A. Guler Yigitoglu, R.T. Wood, A. Saxena, Status report on regulatory criteria applicable to the use of digital twins, 2022a.
- [104] M. Muhlheim, P. Ramuhalli, A. Huning, A. Guler Yigitoglu, A. Saxena, R.T. Wood, Regulatory requirements, guidance, and review of digital twins, *Oak Ridge National Lab.(ORNL)* (2022). Oak Ridge, TN (United States)
- [105] A. Ammar, H. Nassereddine, N. Abdul Baky, A. Abou Kansour, J. Tannoury, H. Urban, C. Schranz, Digital twins in the construction industry: a perspective of practitioners and building authority, *Front. Built Environ.* 8 (2022) 834671.
- [106] B. Lin, Y. Xie, Driving green technology innovation in renewable energy: does venture capital matter? *IEEE Trans. Eng. Manag.* (2023).
- [107] J. Kaur, Fueling healthcare transformation: The nexus of startups, venture capital, and innovation, *In Fostering Innovation in Venture Capital and Startup Ecosystems* (2024) 327–351. IGI Global
- [108] T.S. Messerges, E.A. Dabbish, R.H. Sloan, Investigations of power analysis attacks on smartcards, *Smartcard* 99 (1999) 151–161.
- [109] S.B. Far, M.R. Alagheband, Analysis and improvement of a lightweight anonymous authentication protocol for mobile pay-TV systems, *In 2018 9th International Symposium on Telecommunications (IST)* (2018) 466–473.
- [110] V. Odelu, A.K. Das, A. Goswami, A secure and efficient ECC-based user anonymity preserving single sign-on scheme for distributed computer networks, *Secur. Commun. Netw.* 8 (9) (2015) 1732–1751.
- [111] A.K. Das, N.R. Paul, L. Tripathy, Cryptanalysis and improvement of an access control in user hierarchy based on elliptic curve cryptosystem, *Inf. Sci.* 209 (2012) 80–92.
- [112] S. Banaeian Far, A. Imani Rad, M. Rajabzade Asaar, BB-CSP: an efficient blockchain-based collective salary payment framework using weighted functional encryption, *SN Comput. Sci.* 3 (5) (2022) 408.
- [113] M. Burrows, M. Abadi, R. Needham, A logic of authentication, *ACM Trans. Comput. Syst. (TOCS)* 8 (1) (1990) 18–36.
- [114] M.J. Sadri, M. Rajabzadeh Asaar, A lightweight anonymous two-factor authentication protocol for wireless sensor networks in internet of vehicles, *Int. J. Commun. Syst.* 33 (14) (2020) e4511.
- [115] T.A. Team, *Avispa v1.1 user manual*, *Inf. Soc. Technol. Programme* 62 (2006) 112. <http://www.avispa-project.org>. June 2006
- [116] A. Armando, D. Basin, Y. Boichut, Y. Chevalier, L. Compagna, J. Cuéllar, L. Vigneron, The AVISPA tool for the automated validation of internet security protocols and applications, in: *In Computer Aided Verification: 17th International Conference, CAV 2005, Springer Berlin Heidelberg, Edinburgh, Scotland, UK, 2005*, pp. 281–285. Proceedings 17
- [117] S. Meier, B. Schmidt, C. Cremers, D. Basin, The TAMARIN prover for the symbolic analysis of security protocols, in: *In Computer Aided Verification: 25th International Conference, CAV 2013, Saint Petersburg, Russia, 2013*, pp. 696–701. Proceedings 25
- [118] M. Nakabayashi, Y. Okano, Verification method of key-exchange protocols with a small amount of input using tamarin prover, *In Proceedings of the 2021 International Symposium on Advanced Security on Software and Systems* (2021) 43–50.